

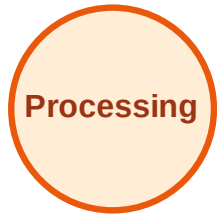
# BFS Traversal

Step 1: Start at A

## Legend



Complete



Processing



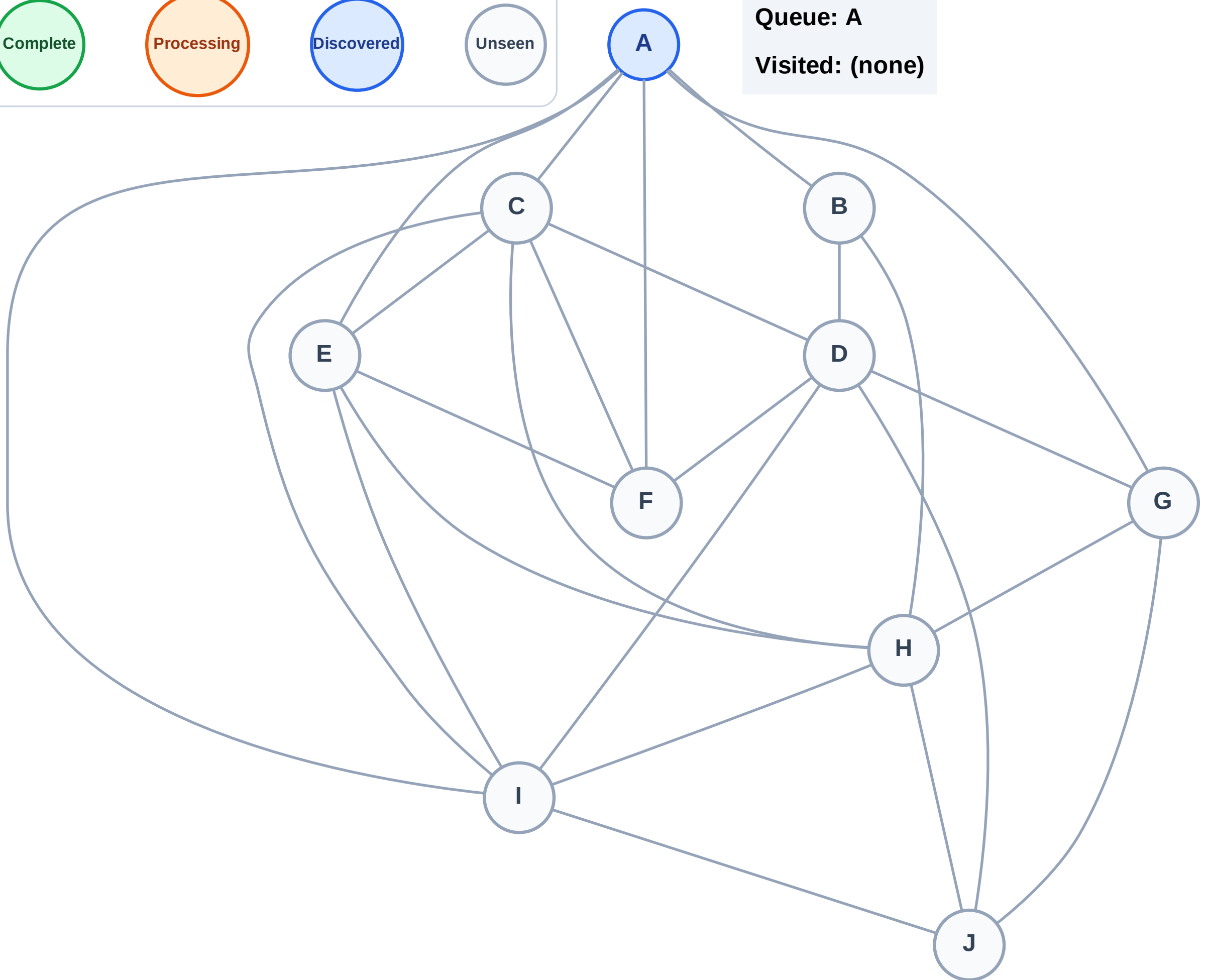
Discovered



Unseen

Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

## Legend



Complete



Processing



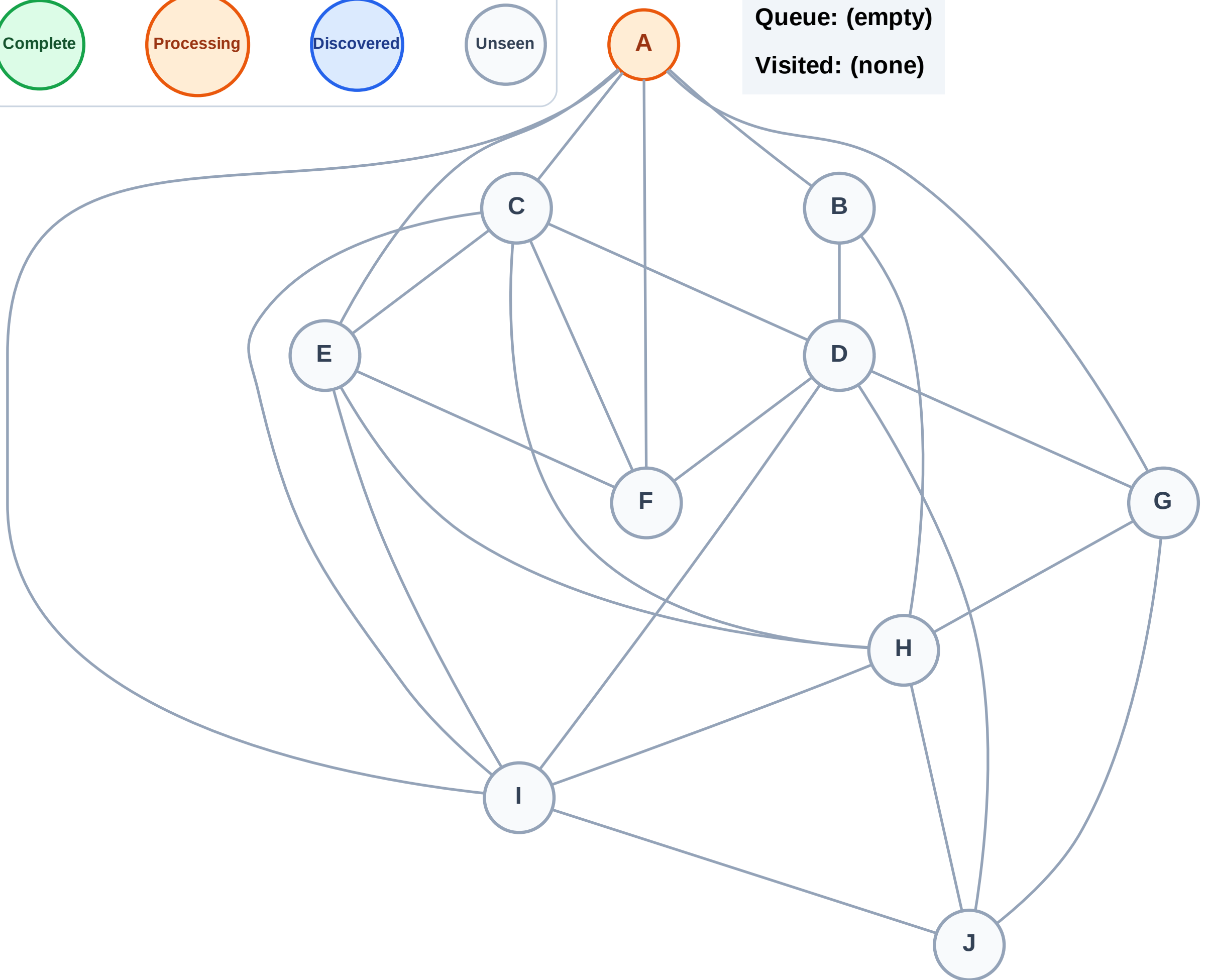
Discovered



Unseen

Queue: (empty)

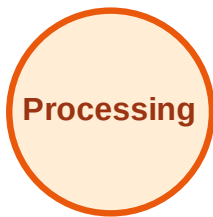
Visited: (none)



# BFS Traversal

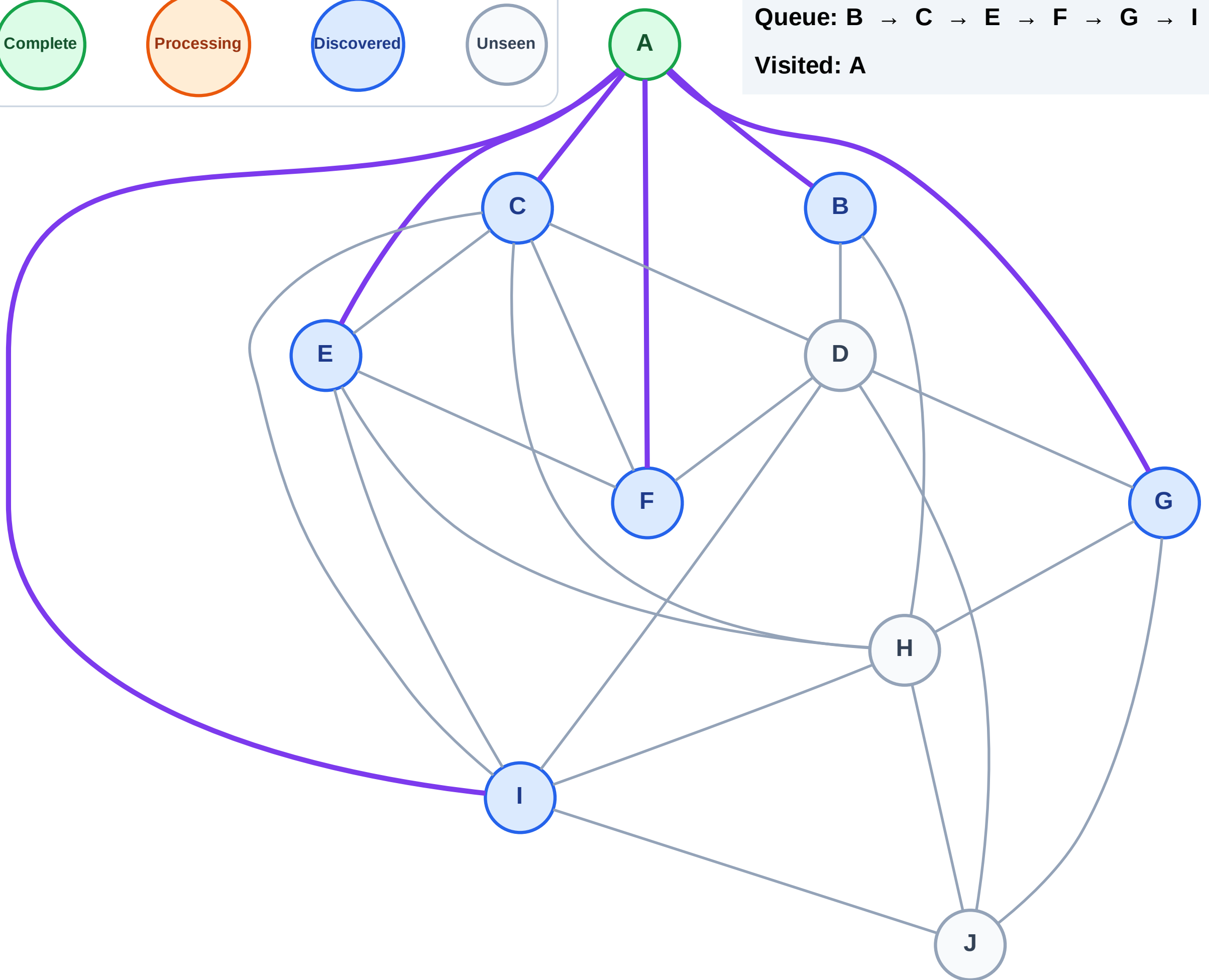
Step 3: Discover B, C, E, F, G, I from A

## Legend



Queue: B → C → E → F → G → I

Visited: A



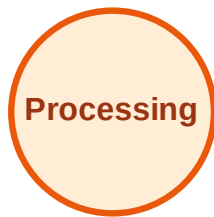
# BFS Traversal

Step 4: Process node B

## Legend



Complete



Processing



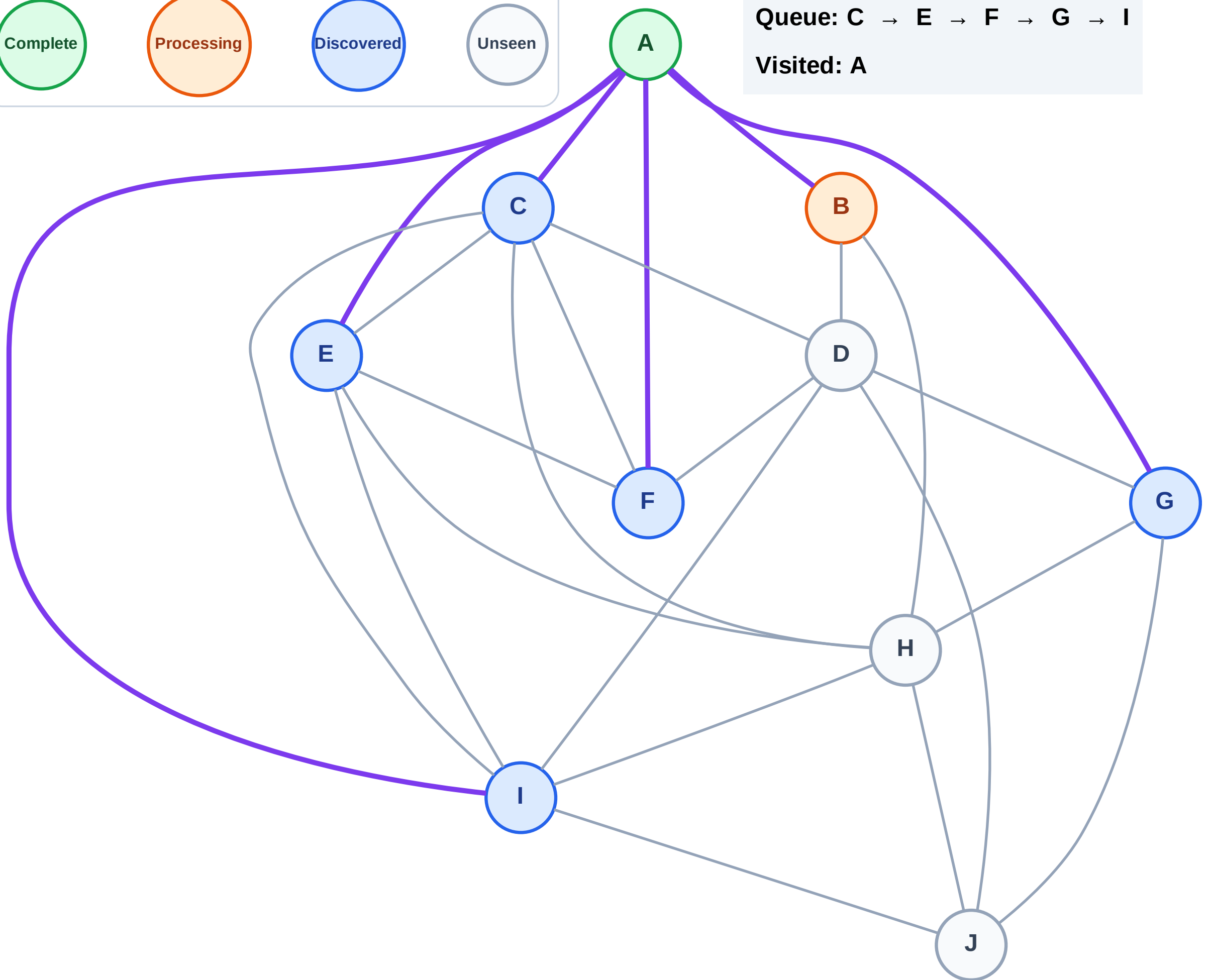
Discovered



Unseen

Queue: C → E → F → G → I

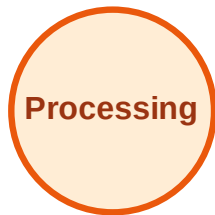
Visited: A



# BFS Traversal

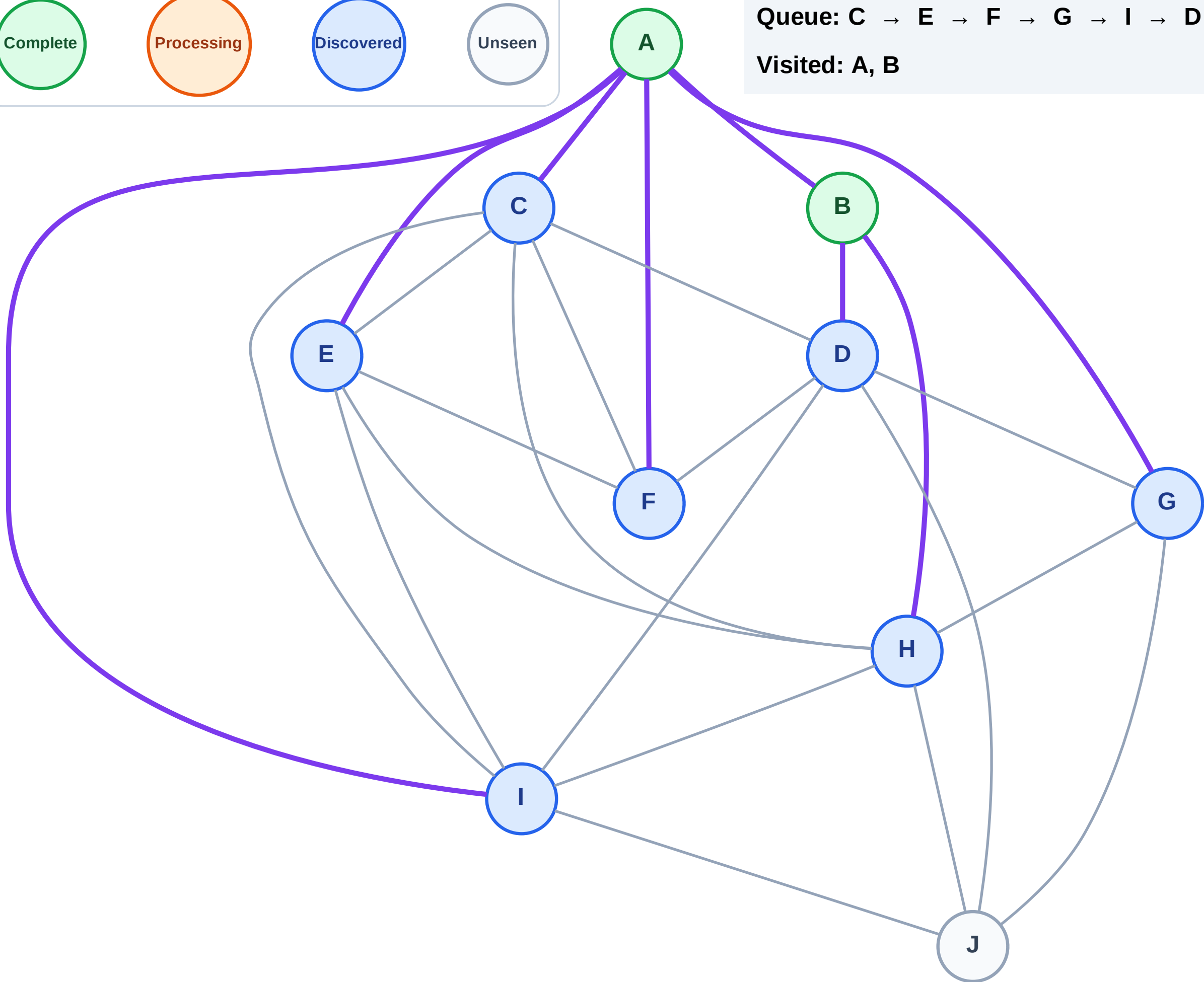
Step 5: Discover D, H from B

## Legend



Queue: C → E → F → G → I → D → H

Visited: A, B



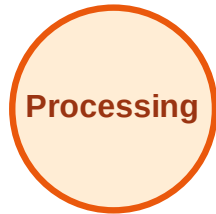
# BFS Traversal

Step 6: Process node C

## Legend



Complete



Processing



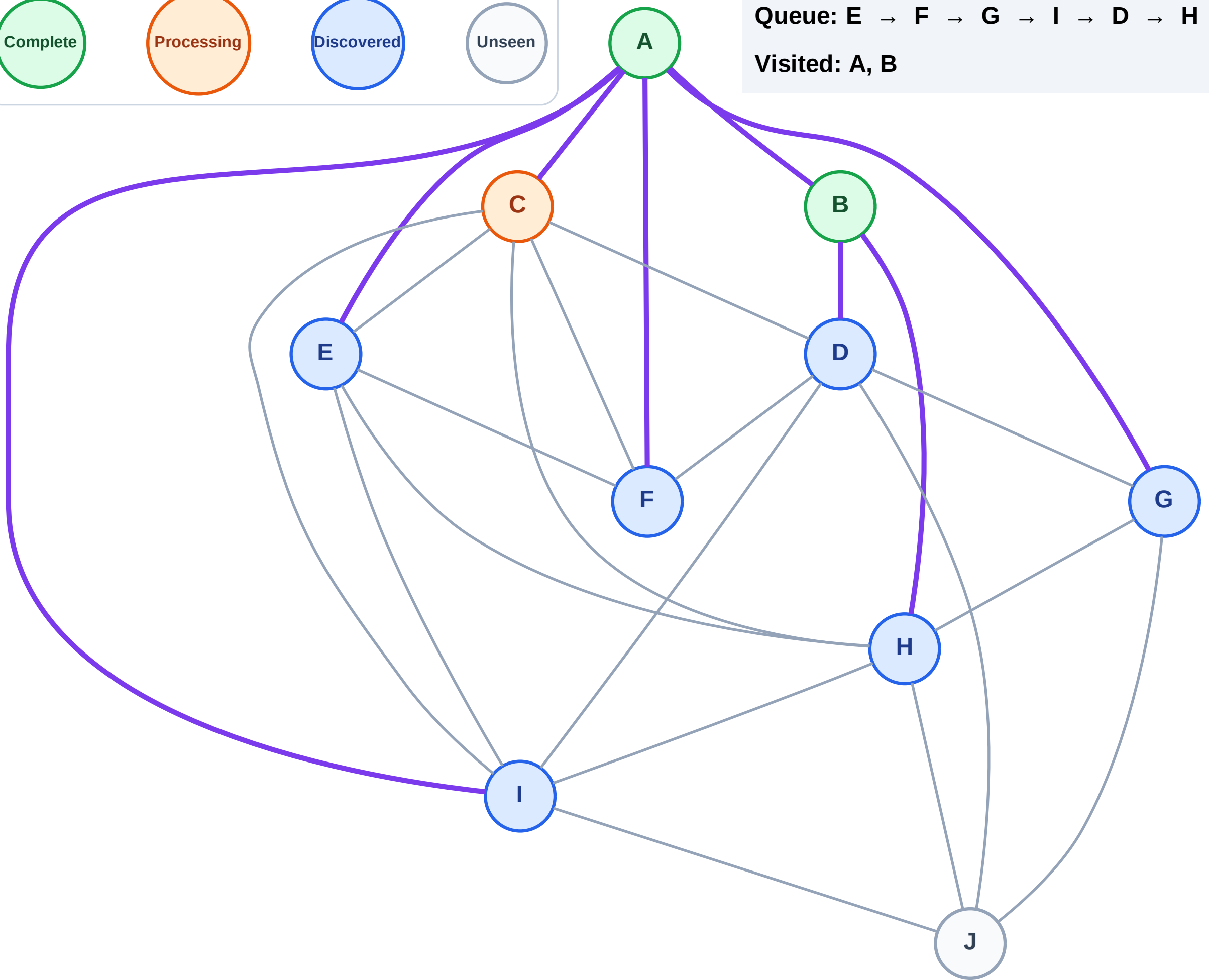
Discovered



Unseen

Queue: E → F → G → I → D → H

Visited: A, B





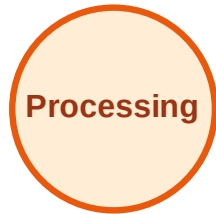
# BFS Traversal

Step 7: No new neighbors from C

## Legend



Complete



Processing



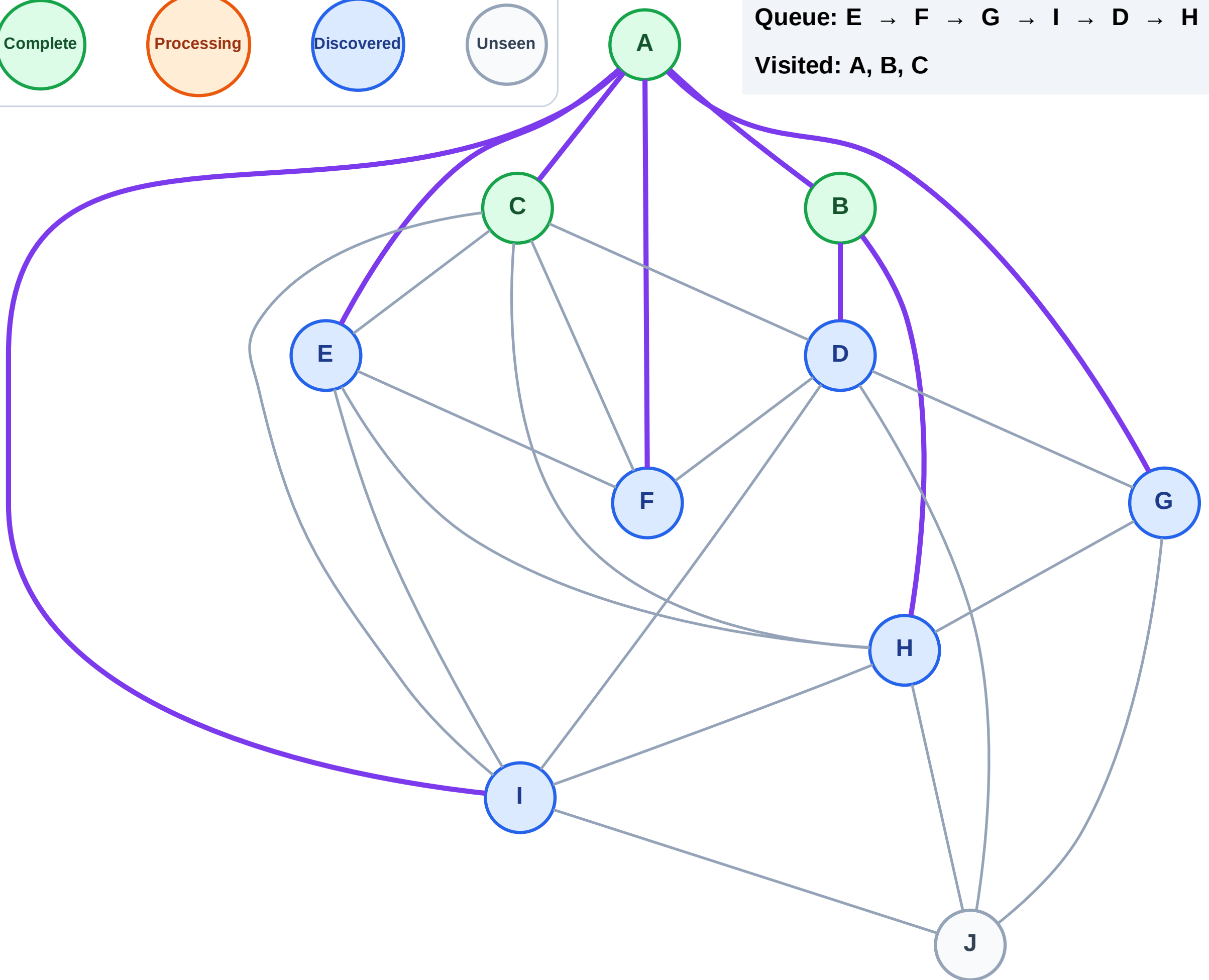
Discovered



Unseen

Queue: E → F → G → I → D → H

Visited: A, B, C



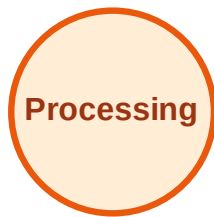
# BFS Traversal

Step 8: Process node E

## Legend



Complete



Processing



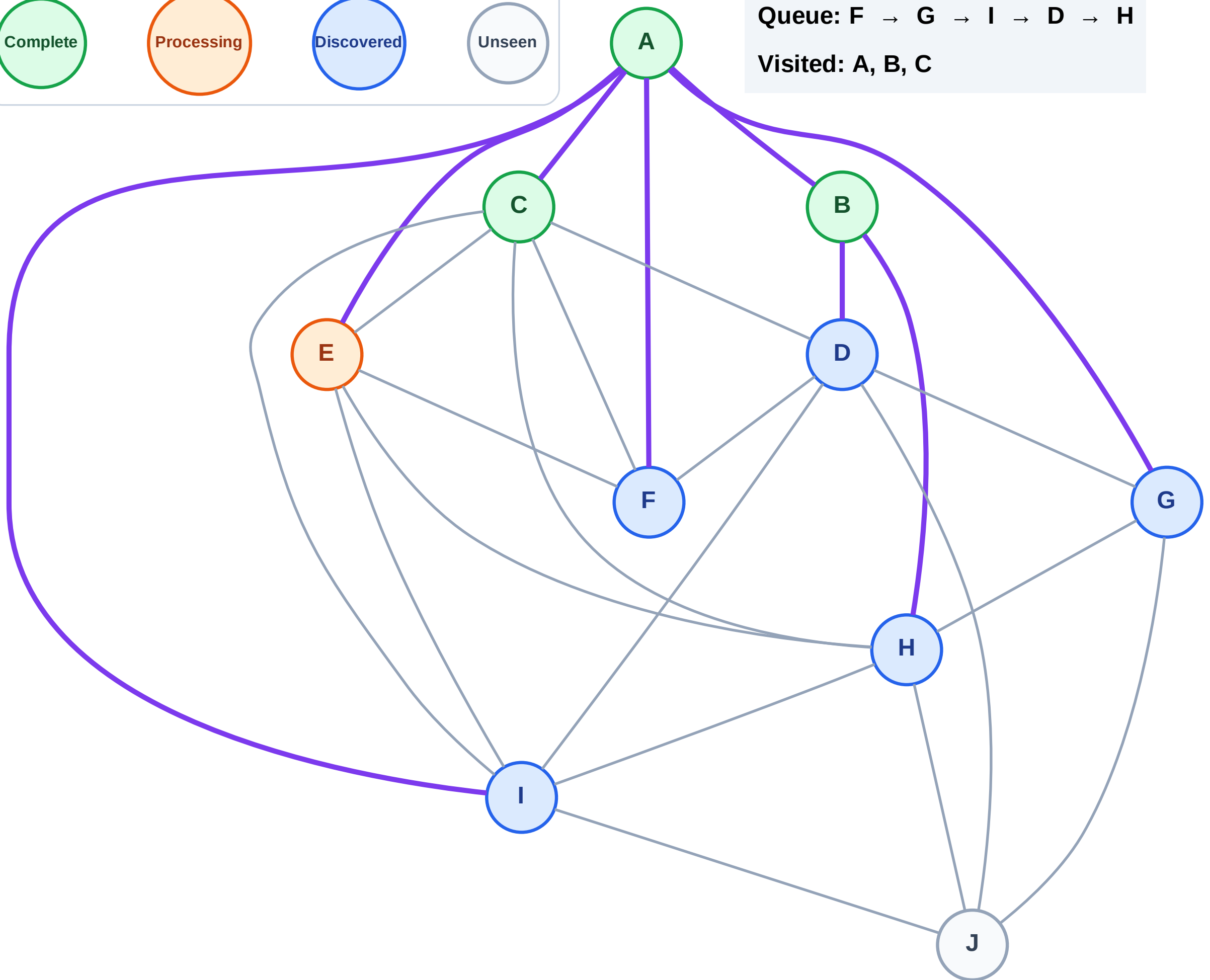
Discovered



Unseen

Queue: F → G → I → D → H

Visited: A, B, C

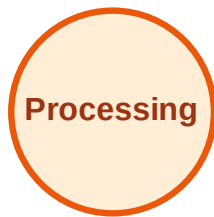




# BFS Traversal

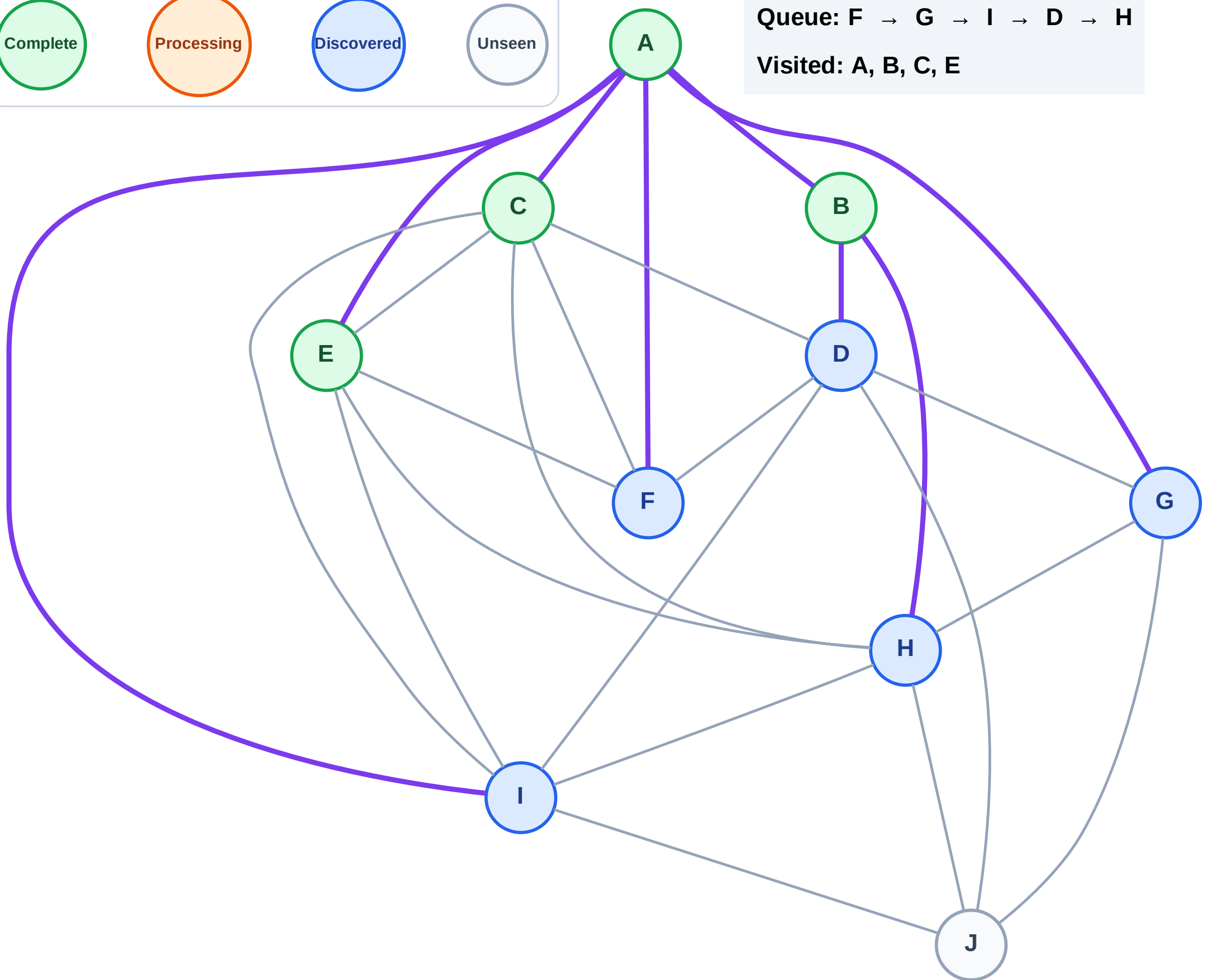
Step 9: No new neighbors from E

## Legend



Queue: F → G → I → D → H

Visited: A, B, C, E



# BFS Traversal

Step 10: Process node F

## Legend

Complete

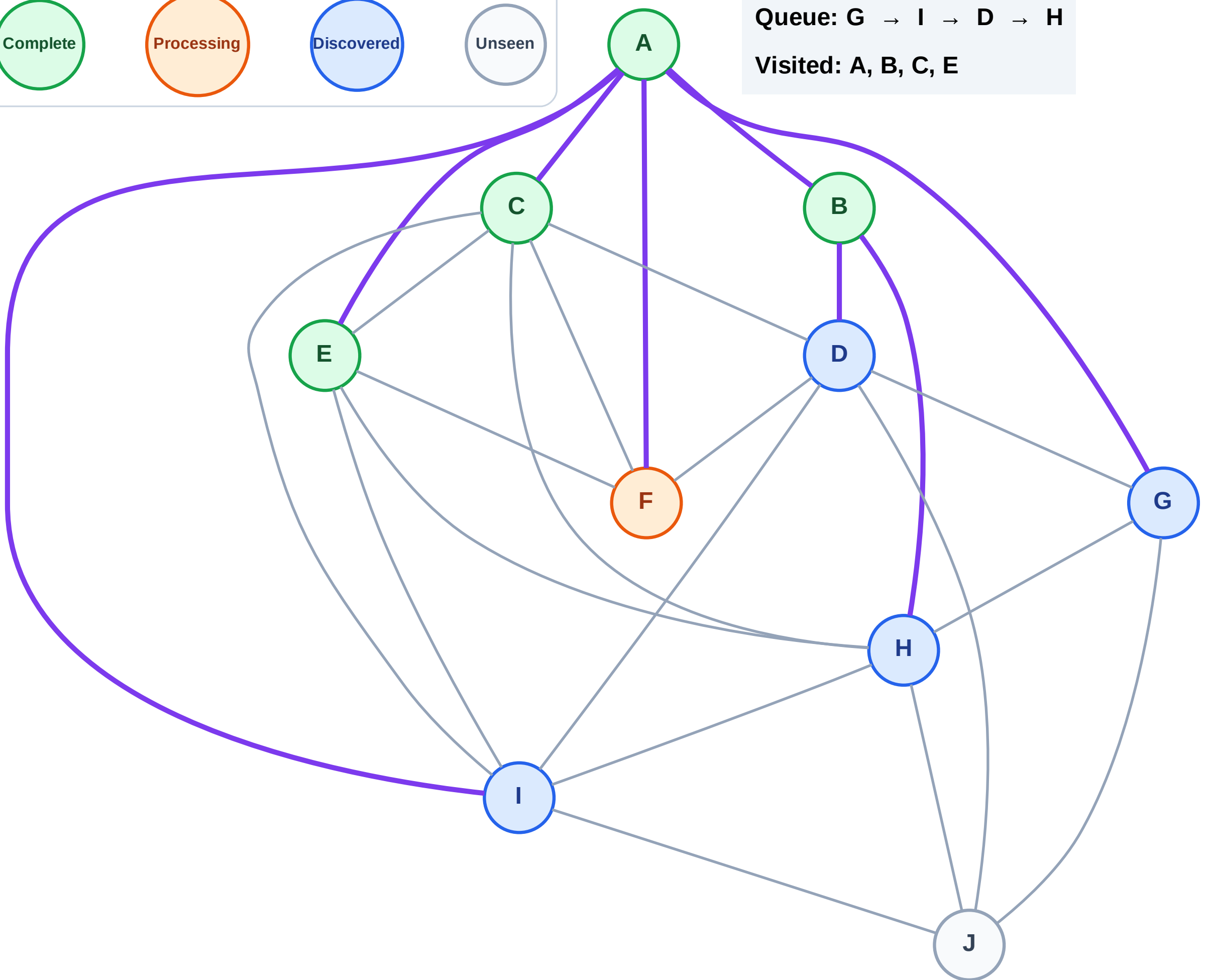
Processing

Discovered

Unseen

Queue: G → I → D → H

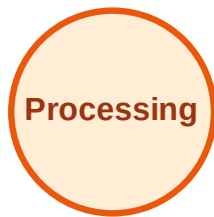
Visited: A, B, C, E



# BFS Traversal

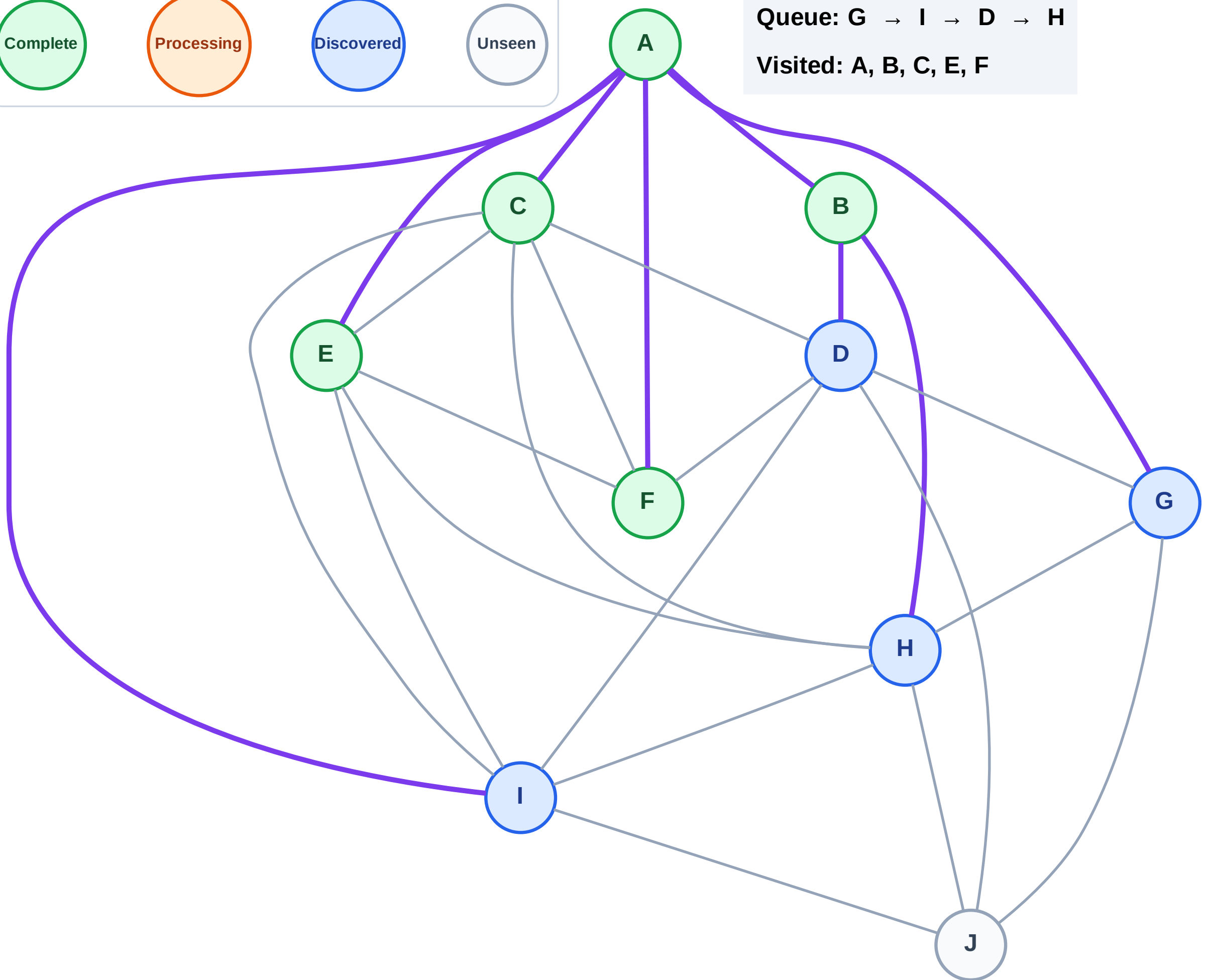
Step 11: No new neighbors from F

## Legend



Queue: G → I → D → H

Visited: A, B, C, E, F



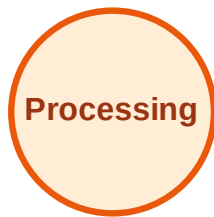
# BFS Traversal

Step 12: Process node G

## Legend



Complete



Processing



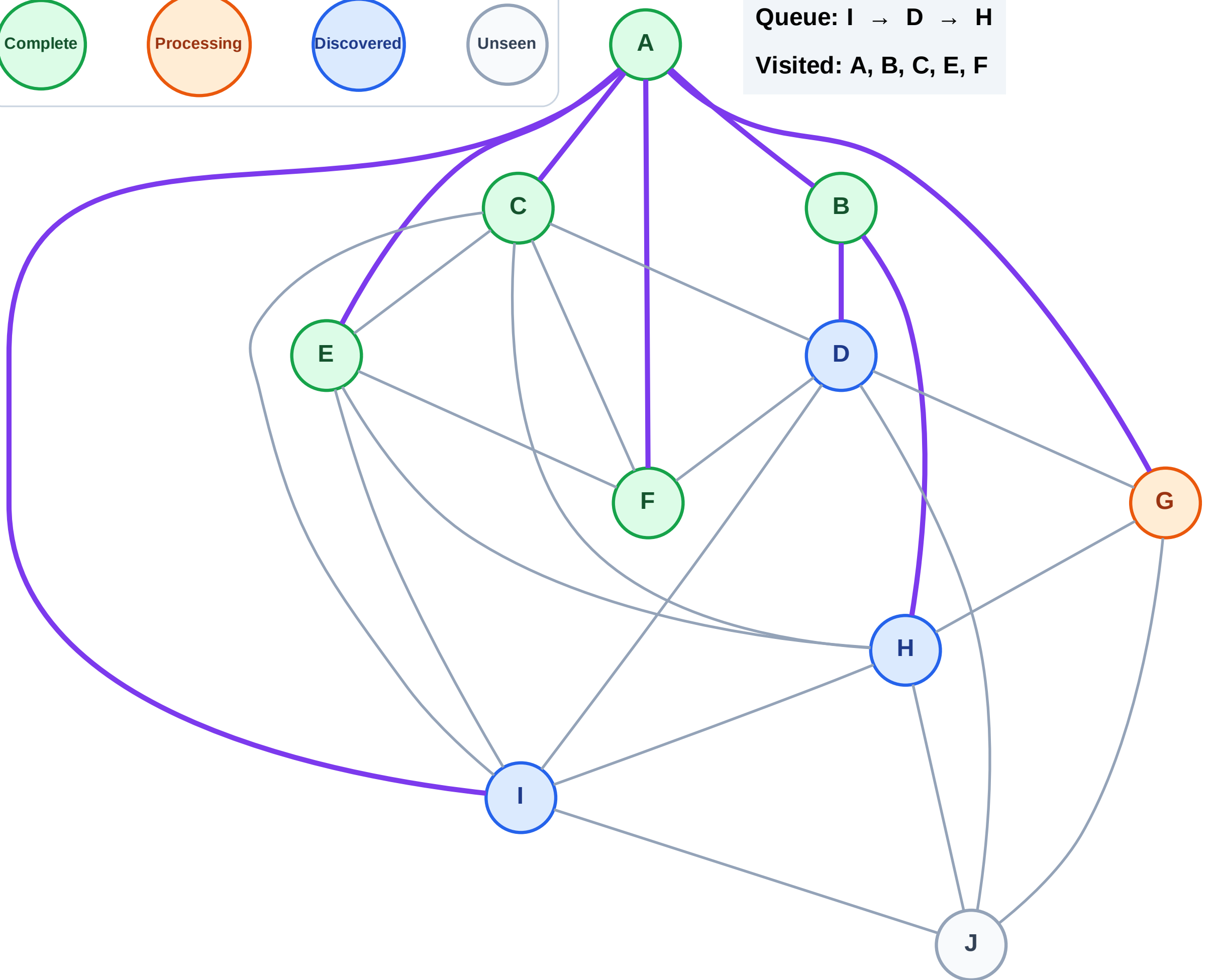
Discovered



Unseen

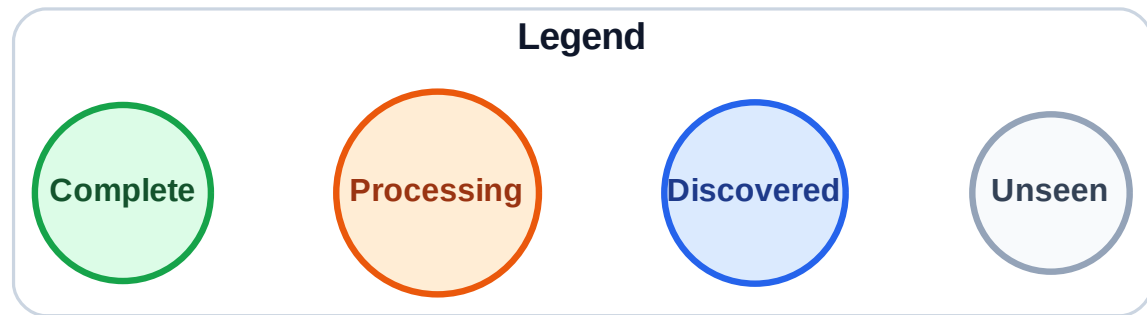
Queue: I → D → H

Visited: A, B, C, E, F

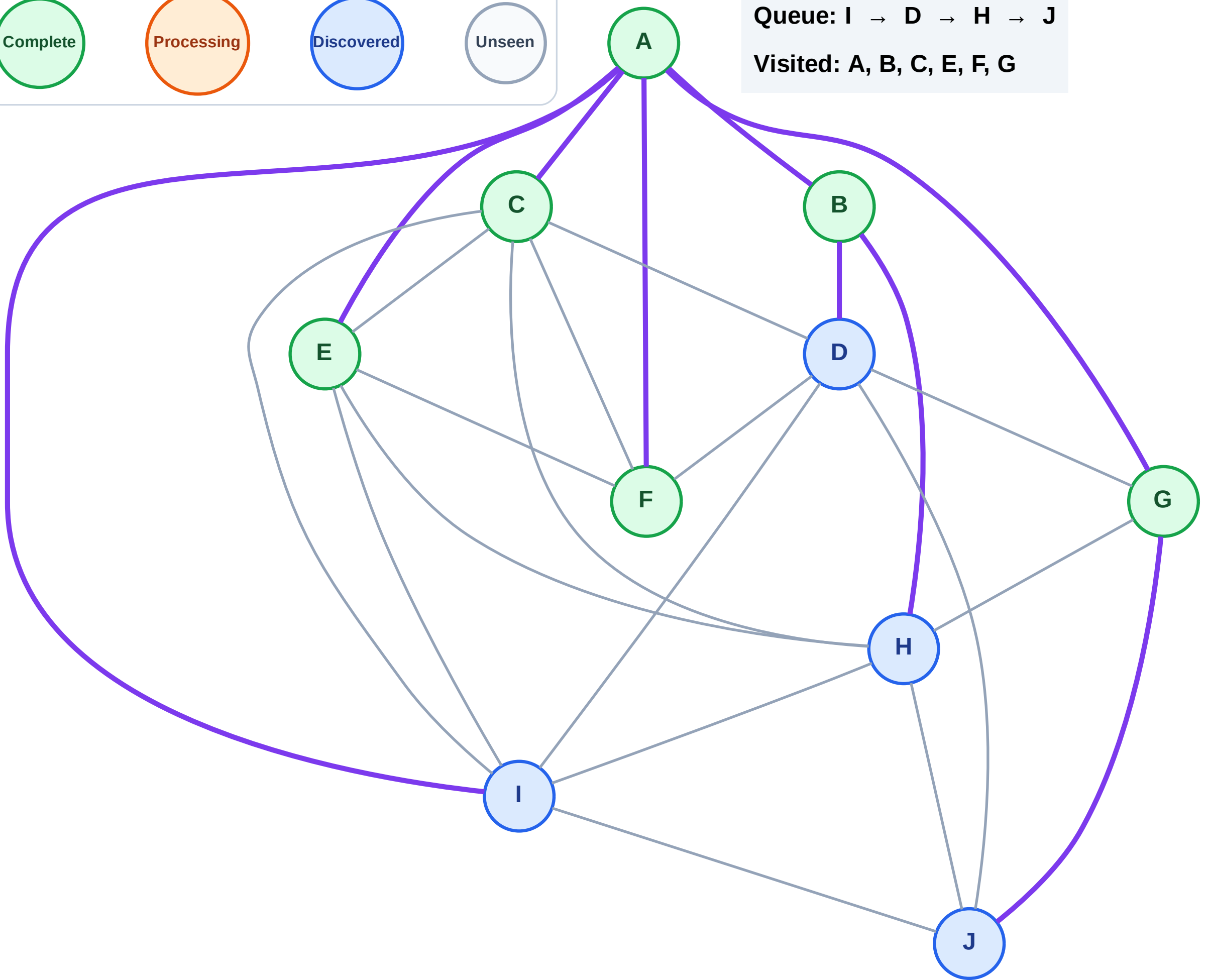


# BFS Traversal

Step 13: Discover J from G



Queue: I → D → H → J  
Visited: A, B, C, E, F, G



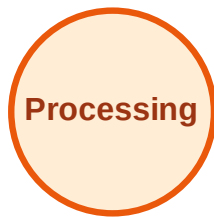
# BFS Traversal

Step 14: Process node I

## Legend



Complete



Processing



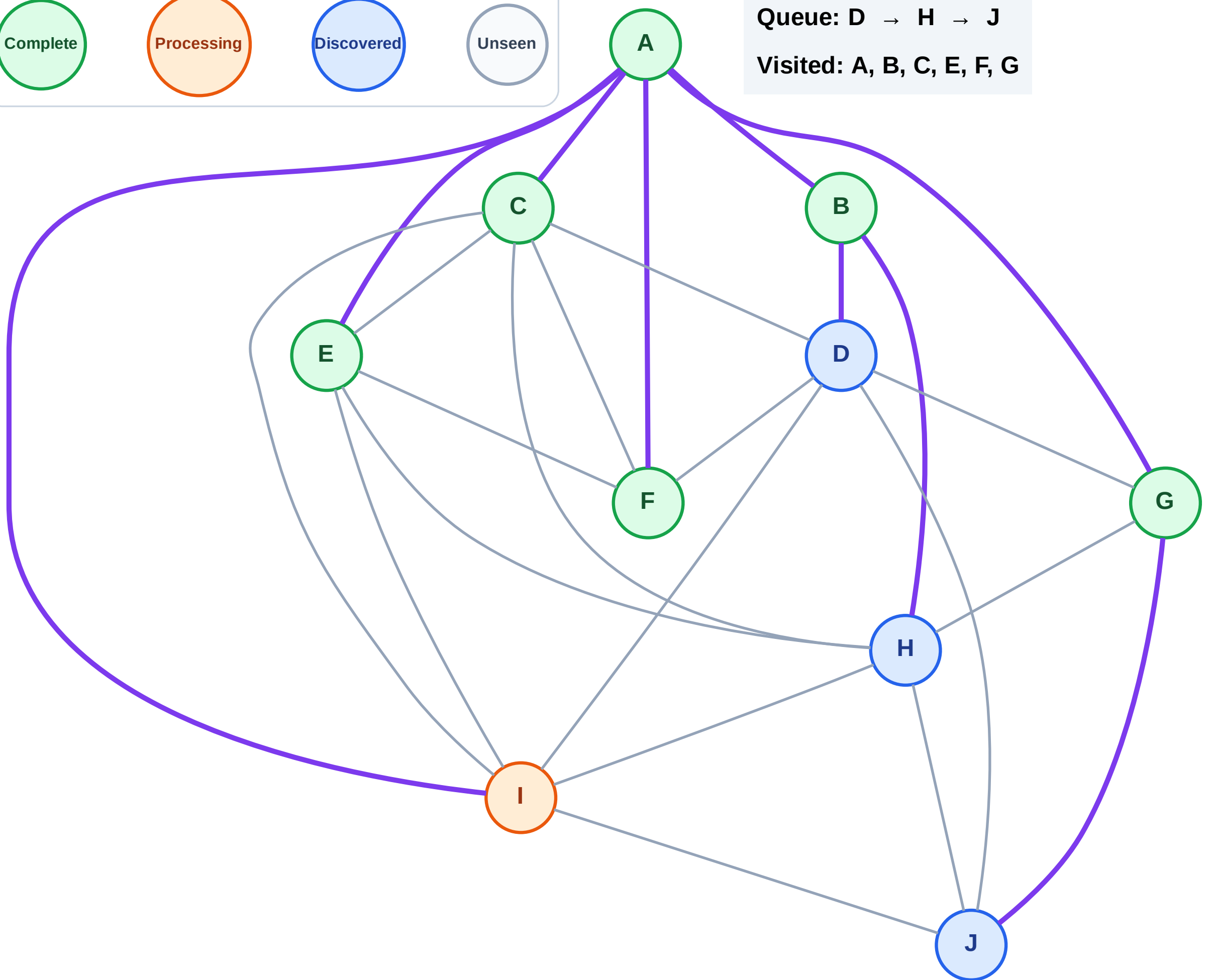
Discovered



Unseen

Queue: D → H → J

Visited: A, B, C, E, F, G

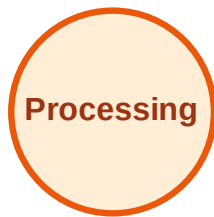




# BFS Traversal

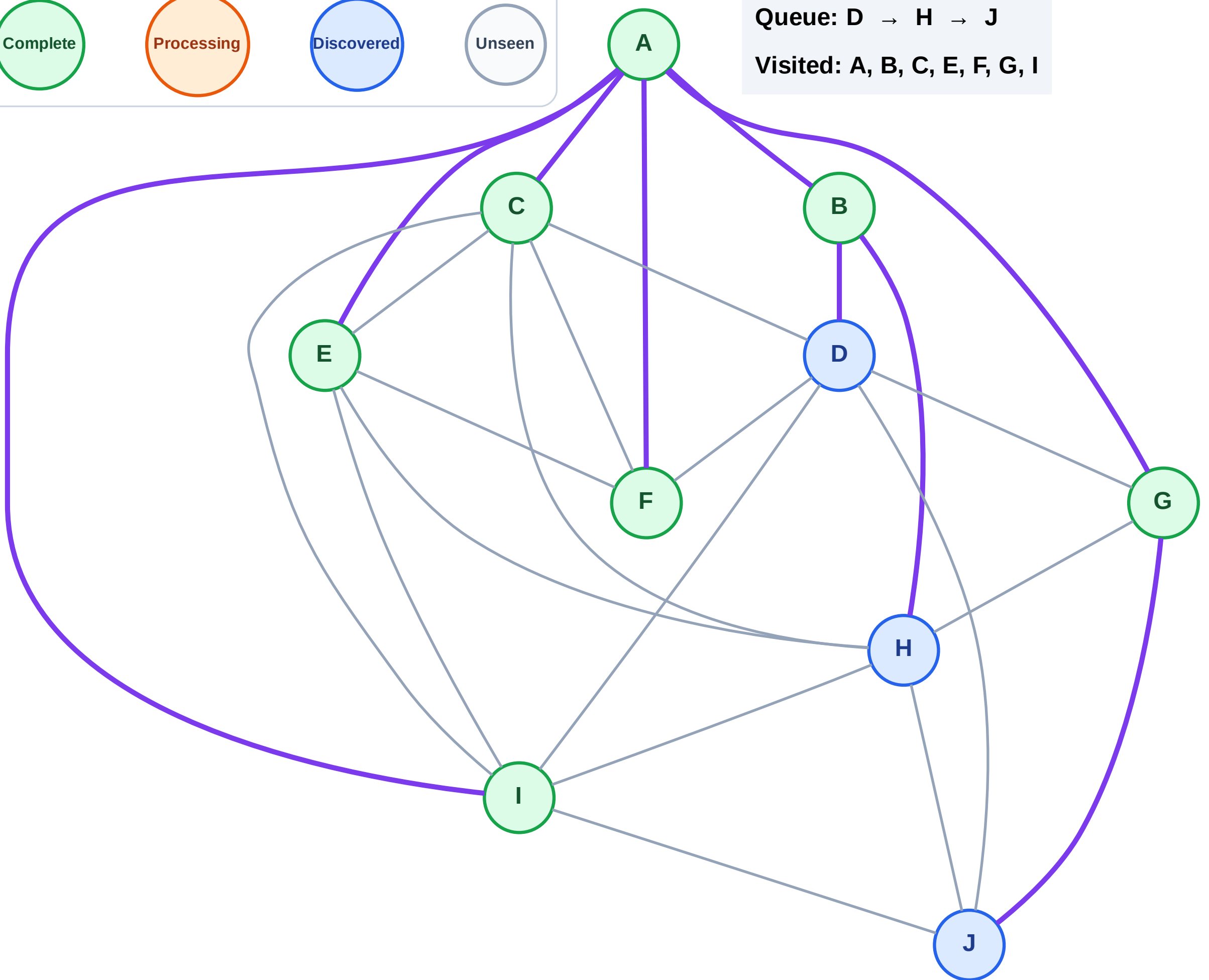
Step 15: No new neighbors from I

## Legend



Queue: D → H → J

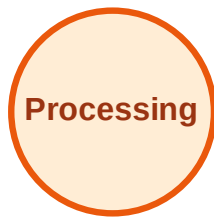
Visited: A, B, C, E, F, G, I



# BFS Traversal

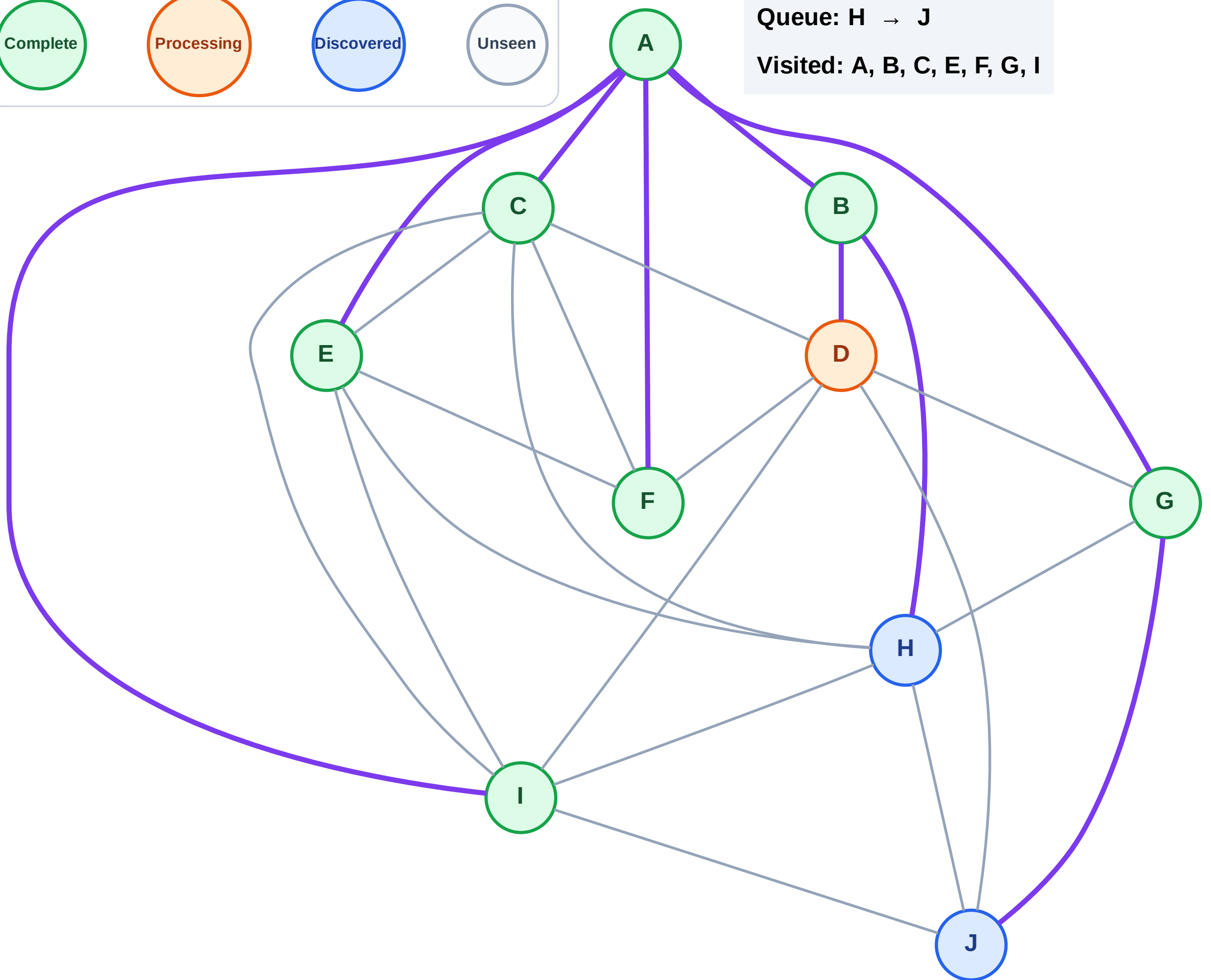
Step 16: Process node D

## Legend



Queue: H → J

Visited: A, B, C, E, F, G, I

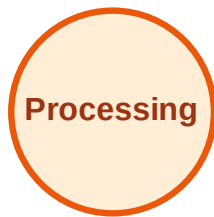




# BFS Traversal

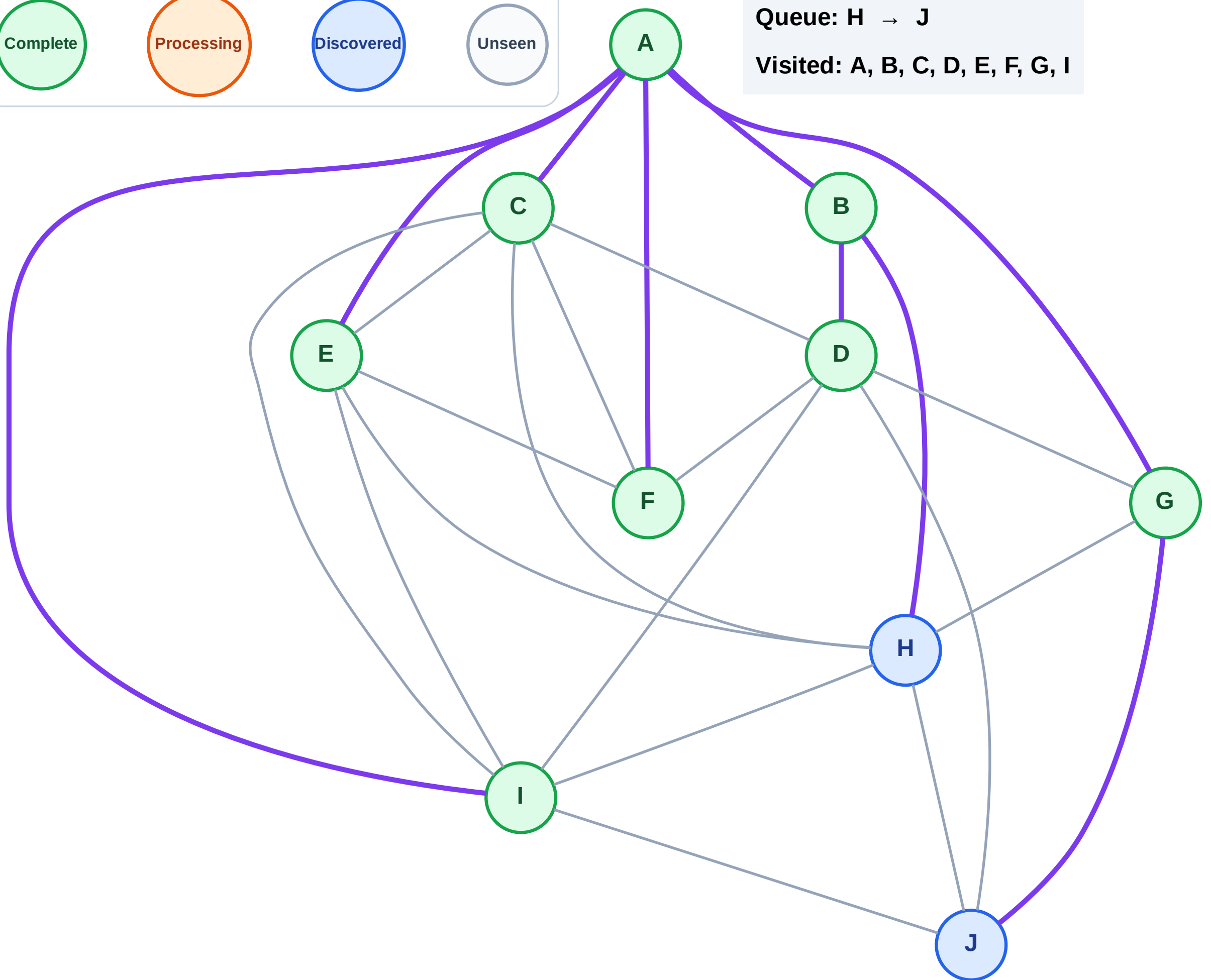
Step 17: No new neighbors from D

## Legend



Queue: H → J

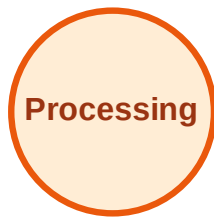
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

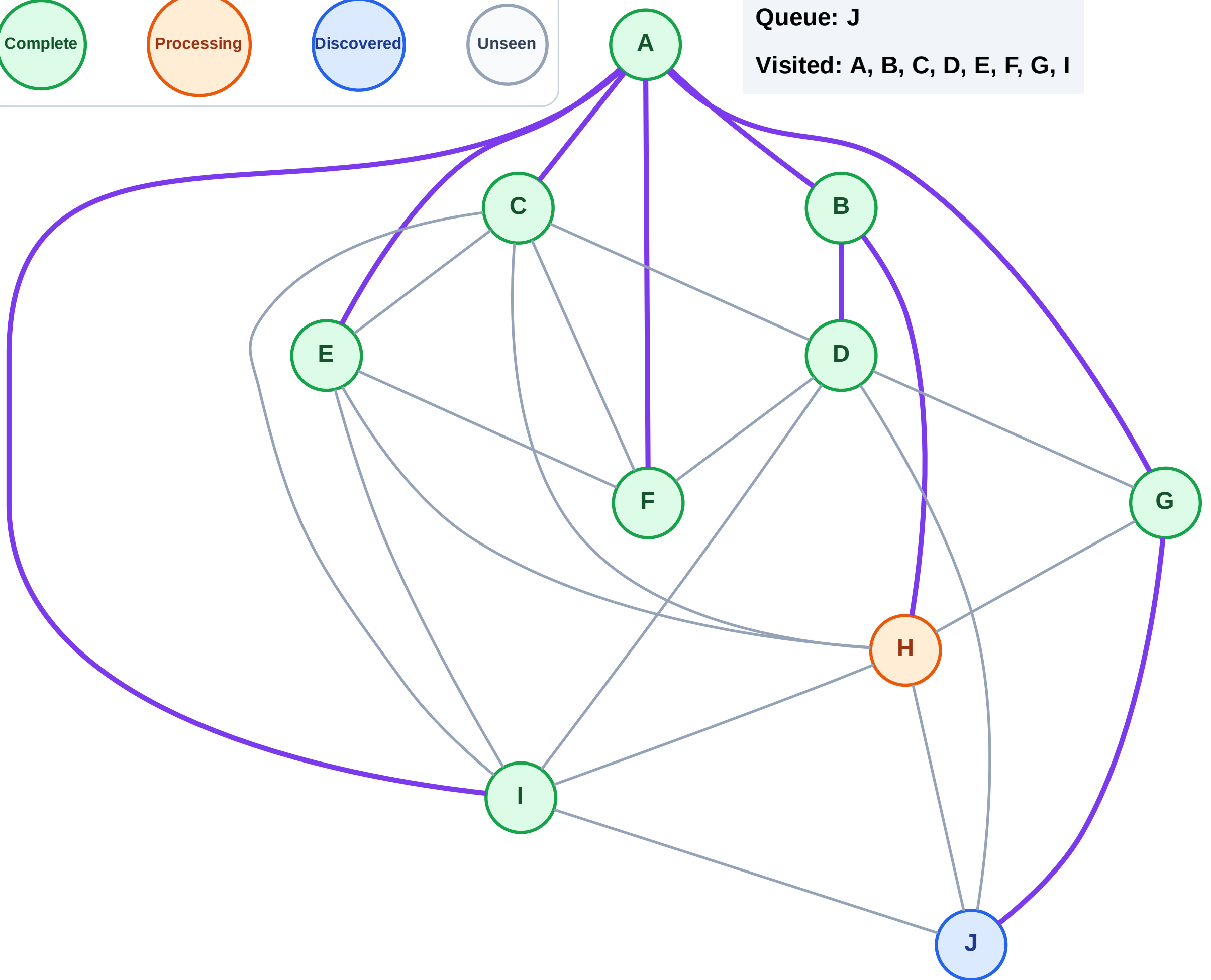
Step 18: Process node H

## Legend



Queue: J

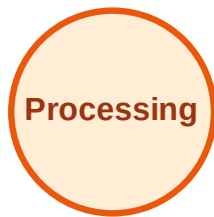
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

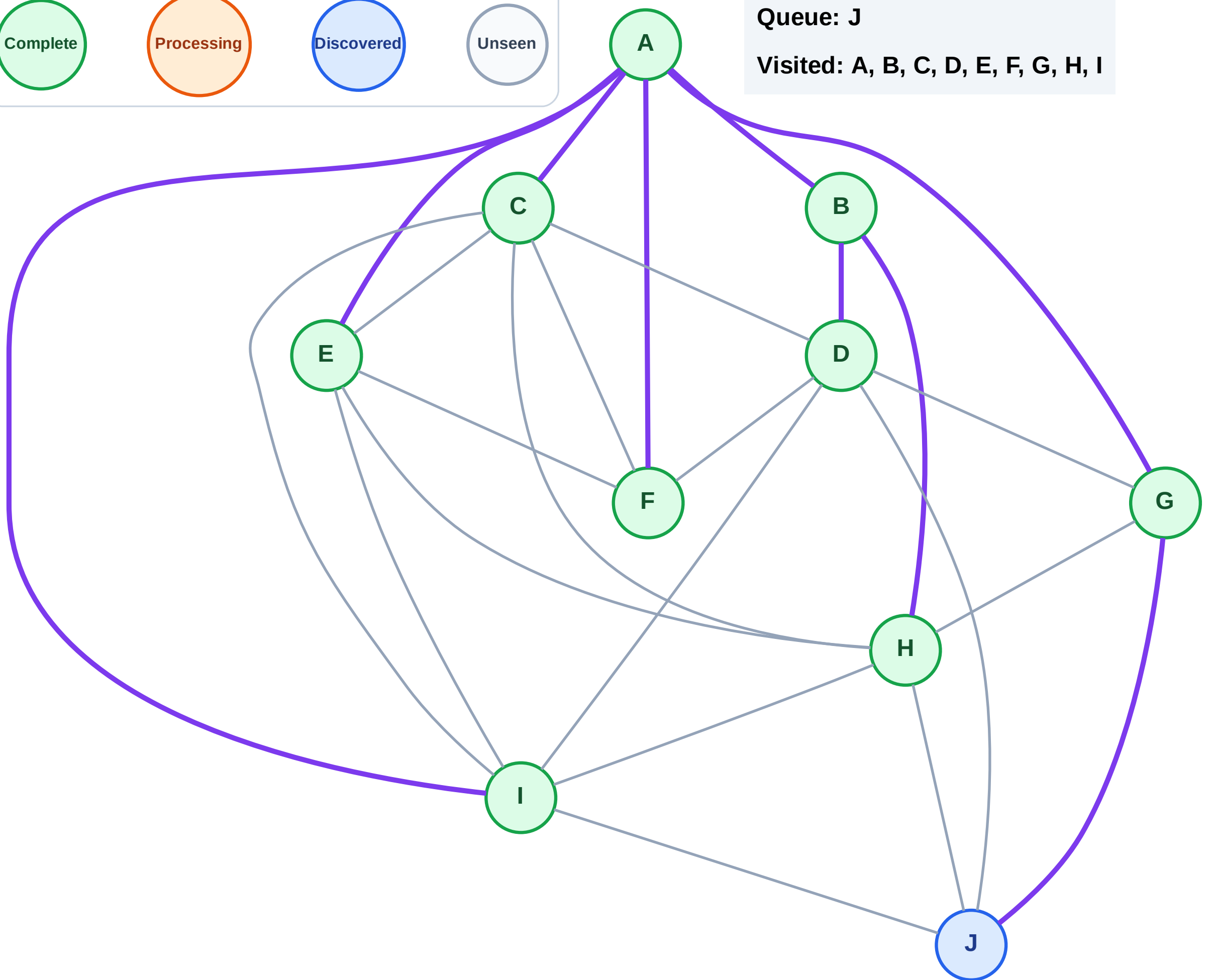
Step 19: No new neighbors from H

## Legend



Queue: J

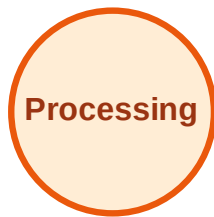
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

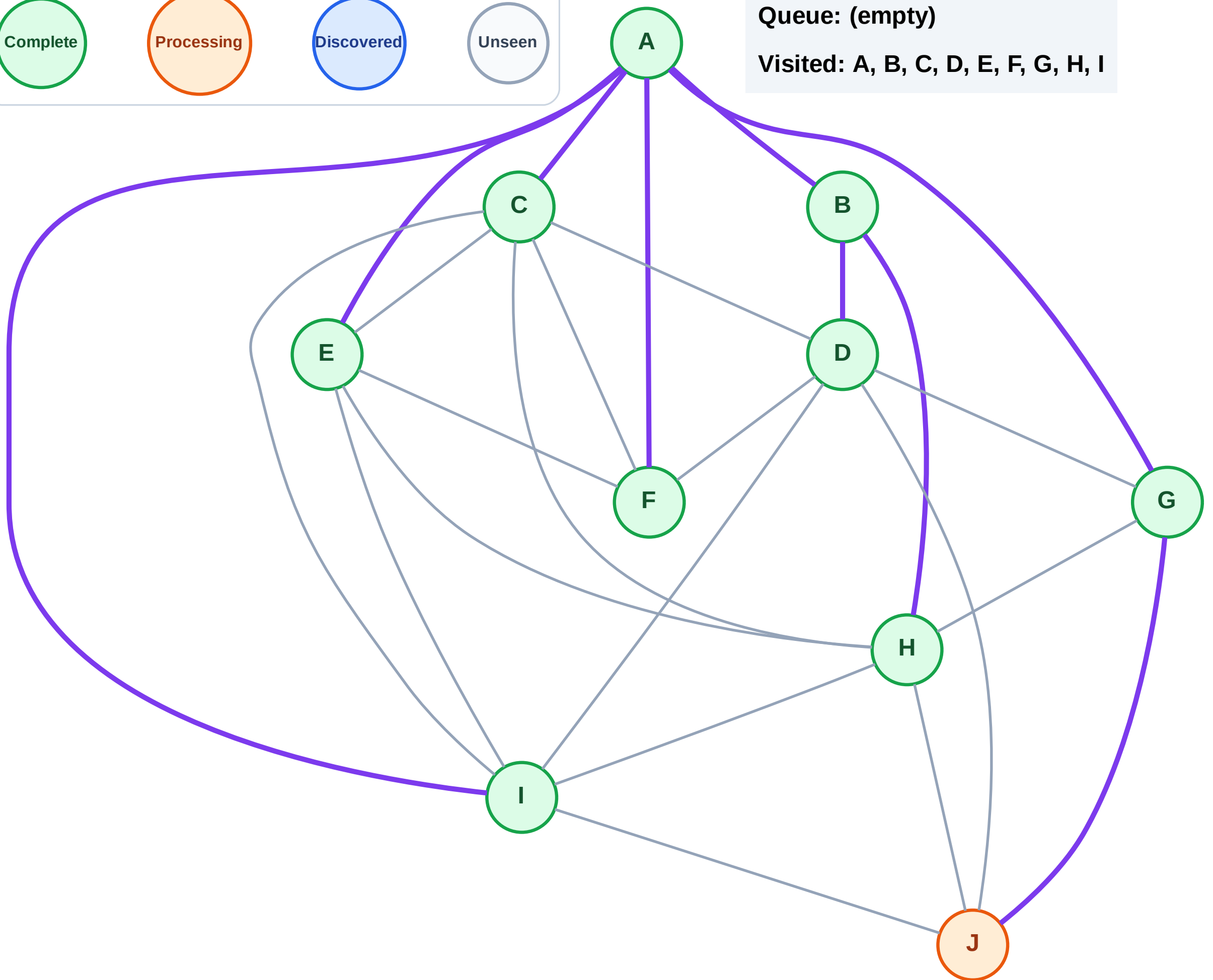
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



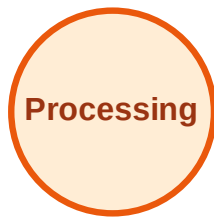
# BFS Traversal

Step 21: No new neighbors from J

## Legend



Complete



Processing



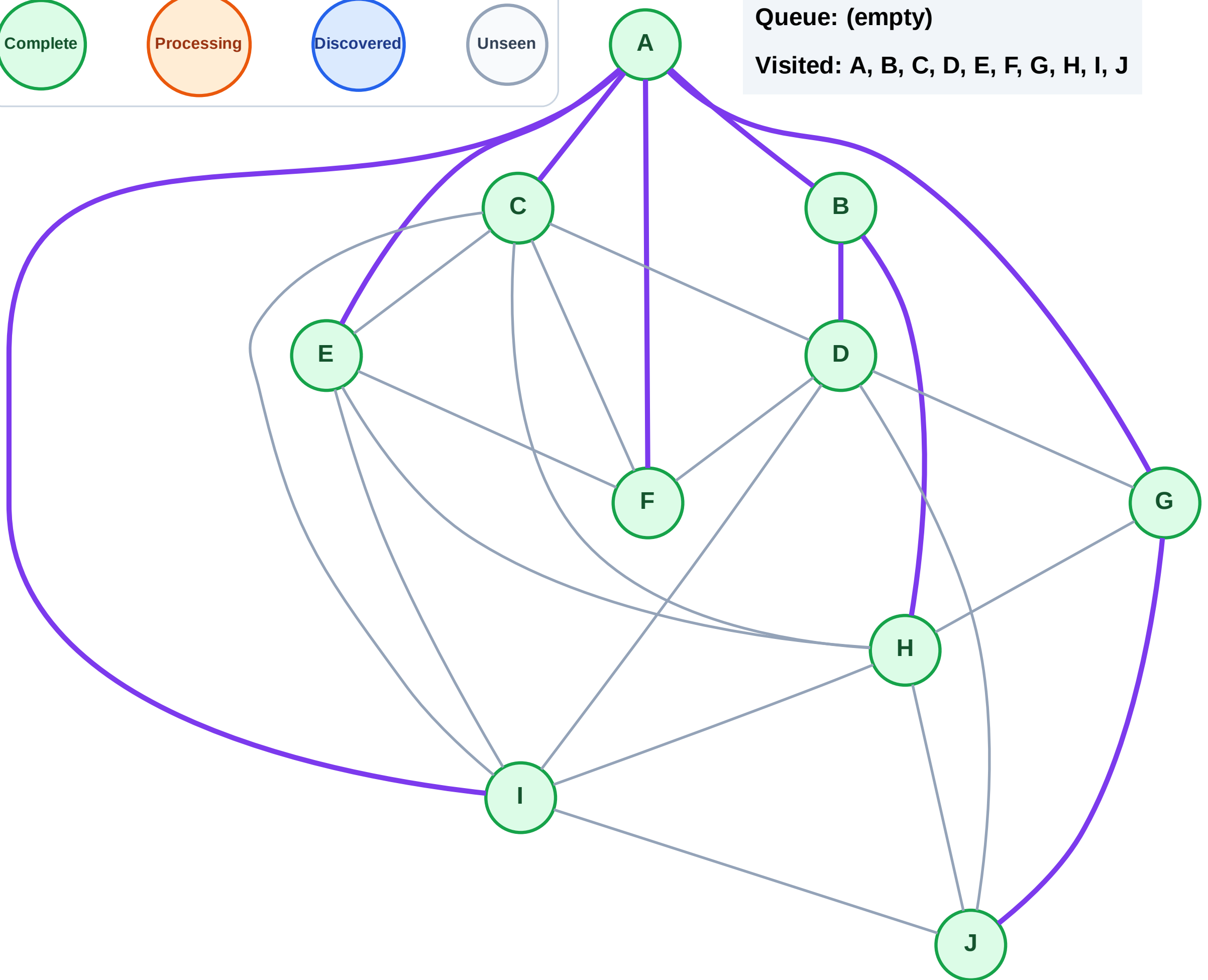
Discovered



Unseen

Queue: (empty)

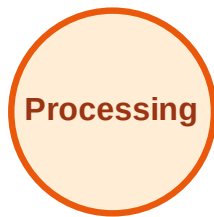
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

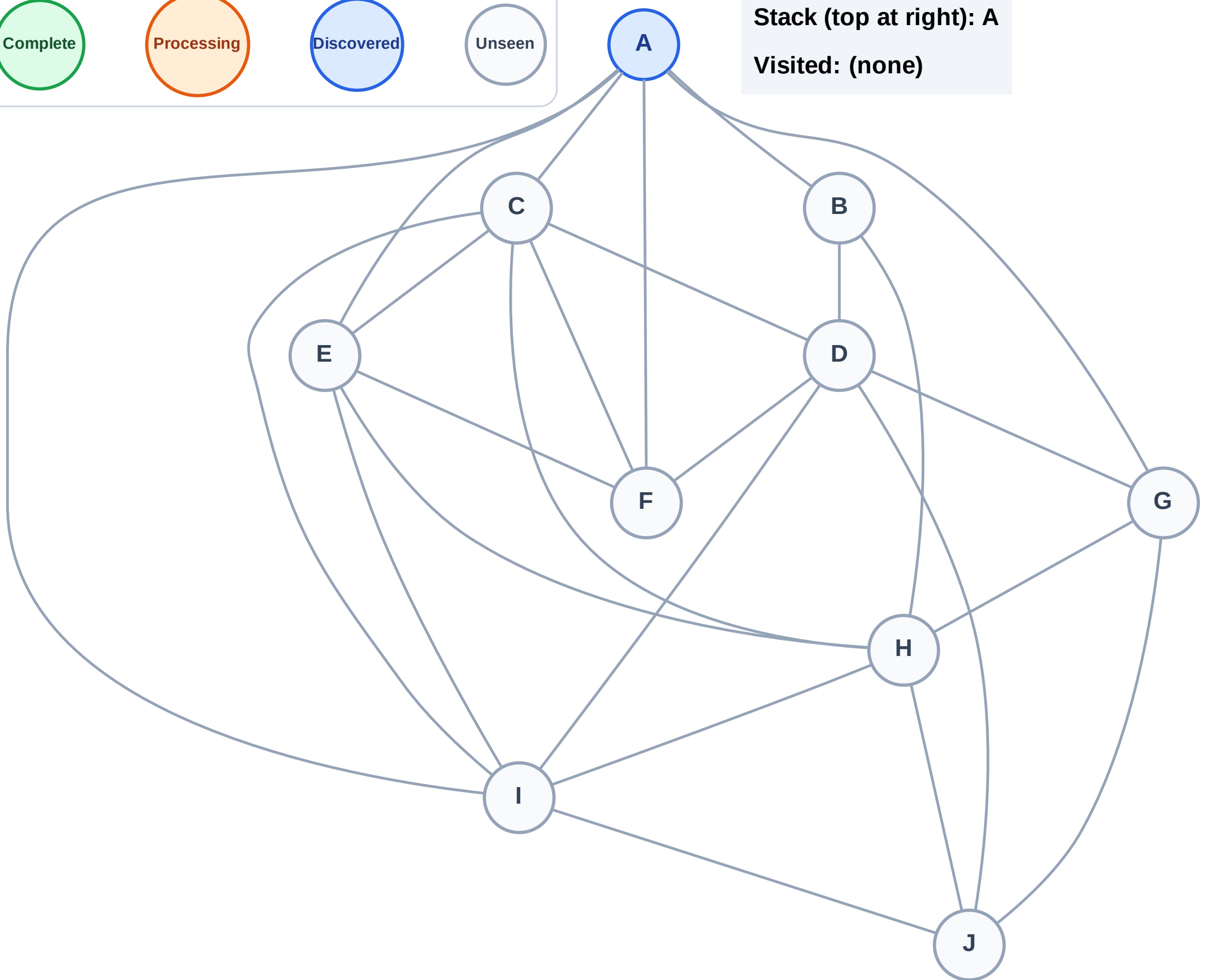
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)

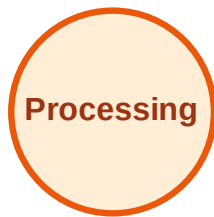




# DFS Traversal

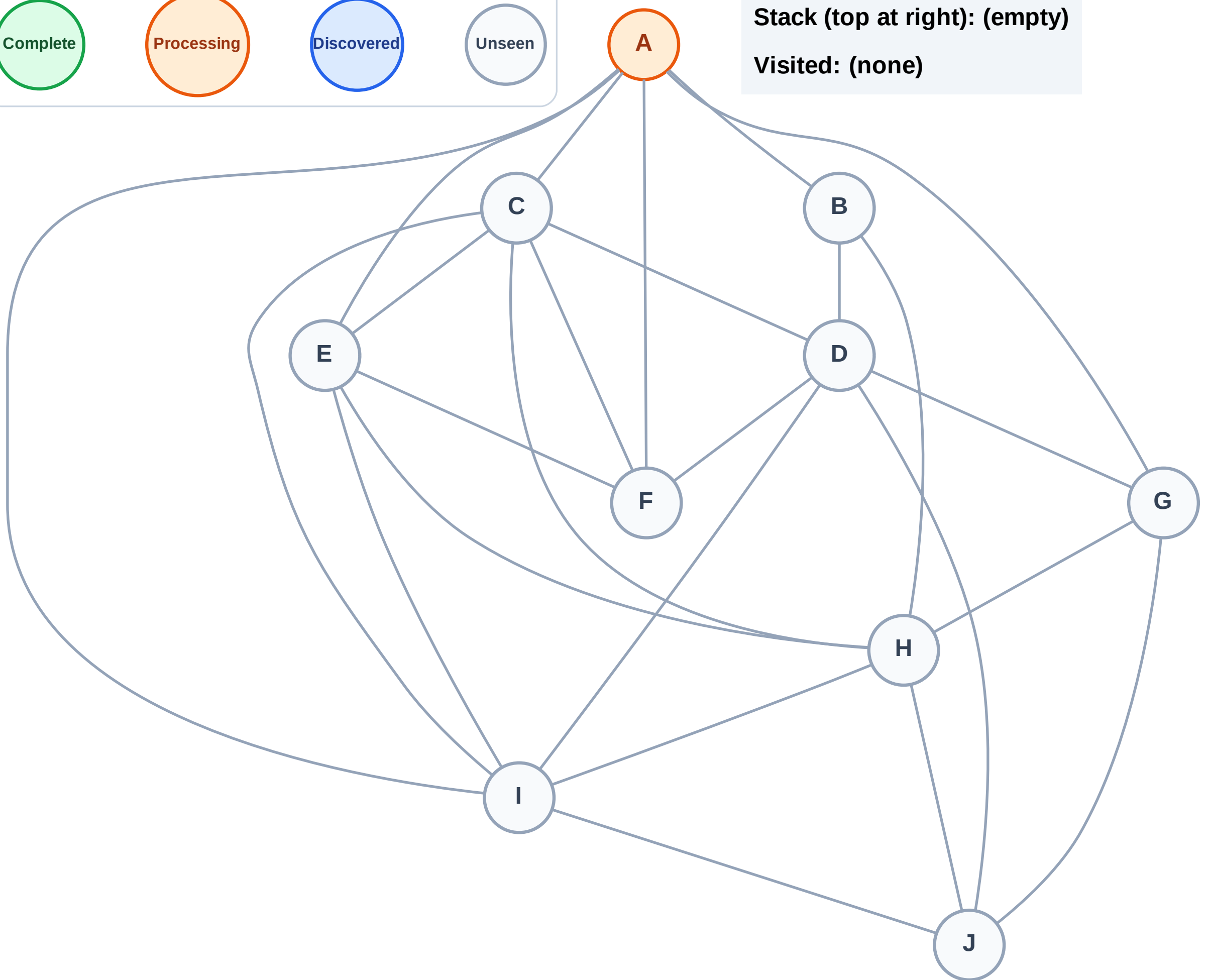
Step 2: Process node A

## Legend



Stack (top at right): (empty)

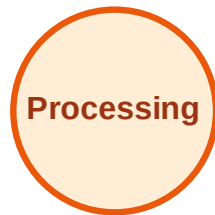
Visited: (none)



# DFS Traversal

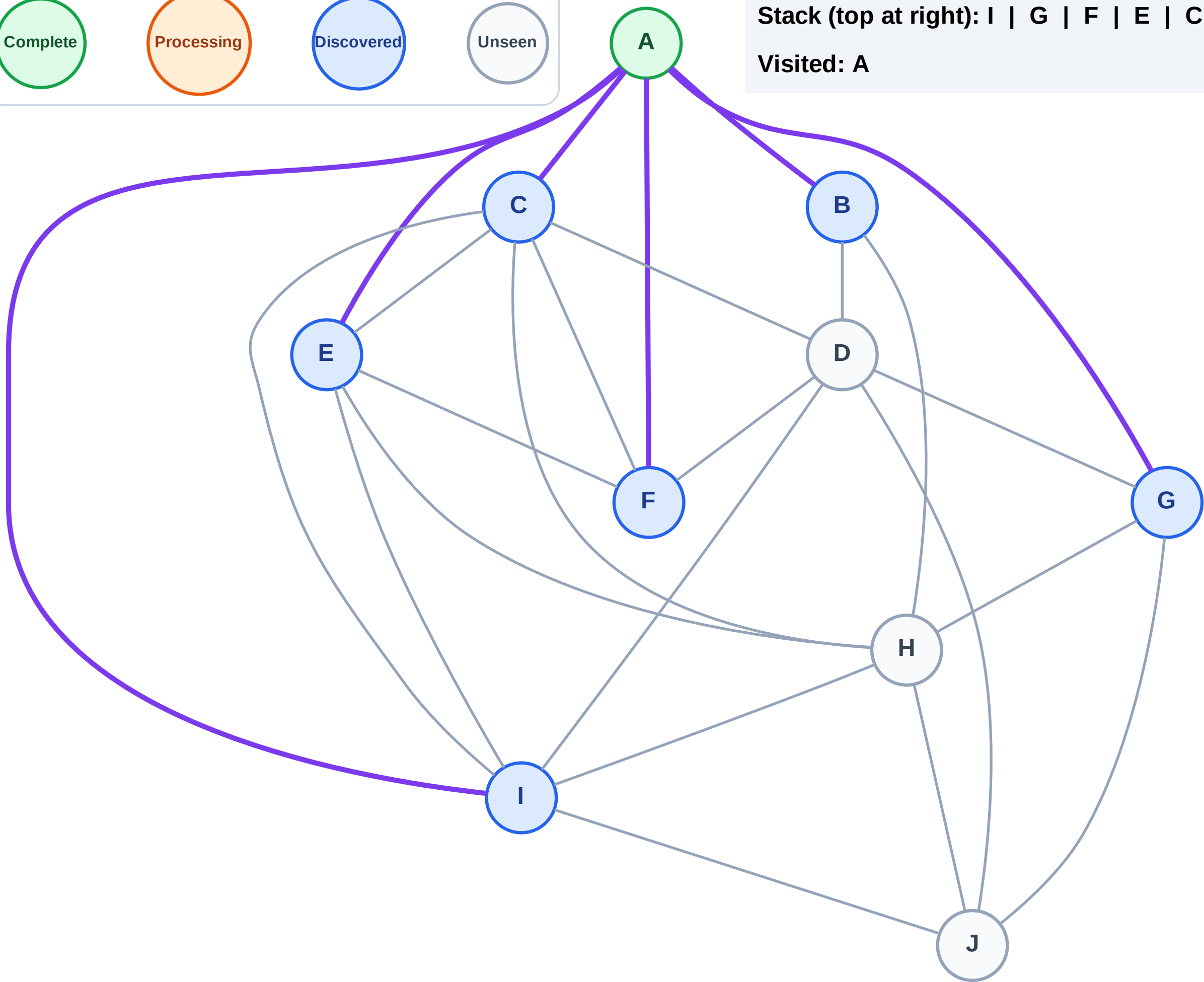
Step 3: Discover I, G, F, E, C, B from A

## Legend



Stack (top at right): I | G | F | E | C | B

Visited: A

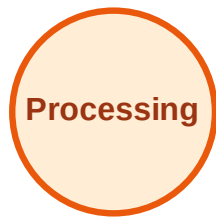




# DFS Traversal

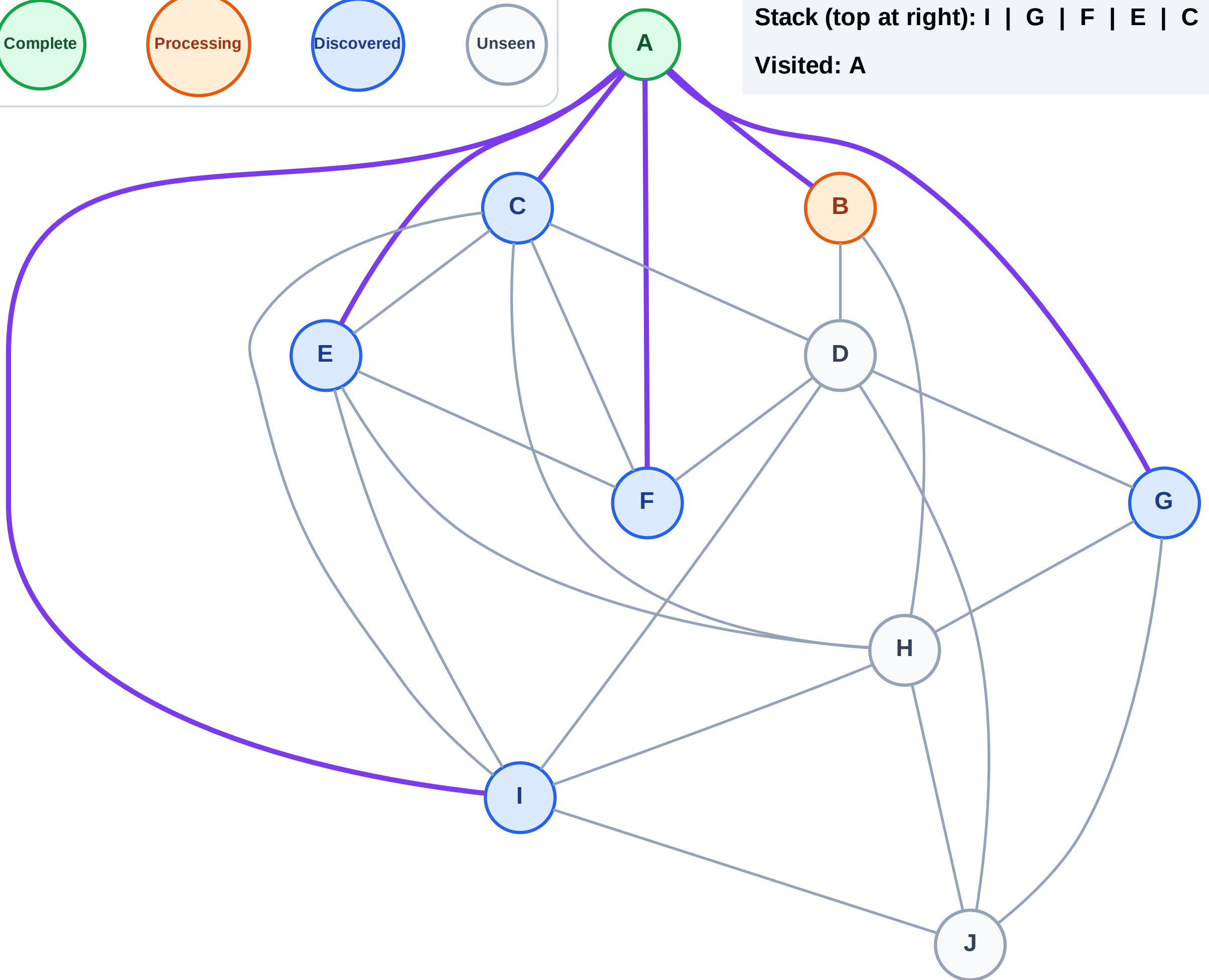
Step 4: Process node B

## Legend



Stack (top at right): I | G | F | E | C

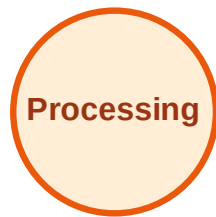
Visited: A



# DFS Traversal

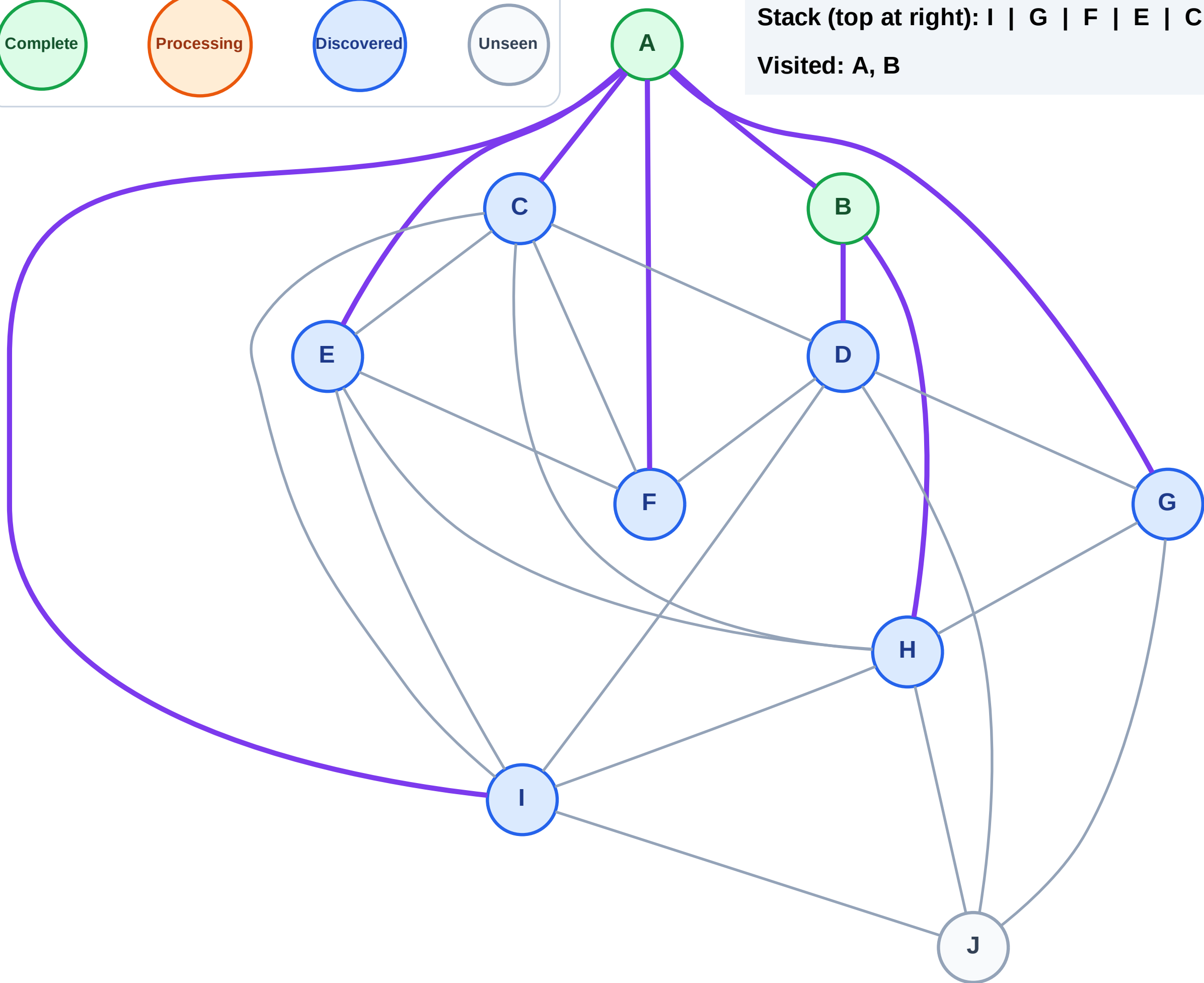
Step 5: Discover H, D from B

## Legend



Stack (top at right): I | G | F | E | C | H | D

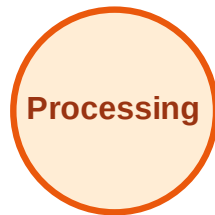
Visited: A, B



# DFS Traversal

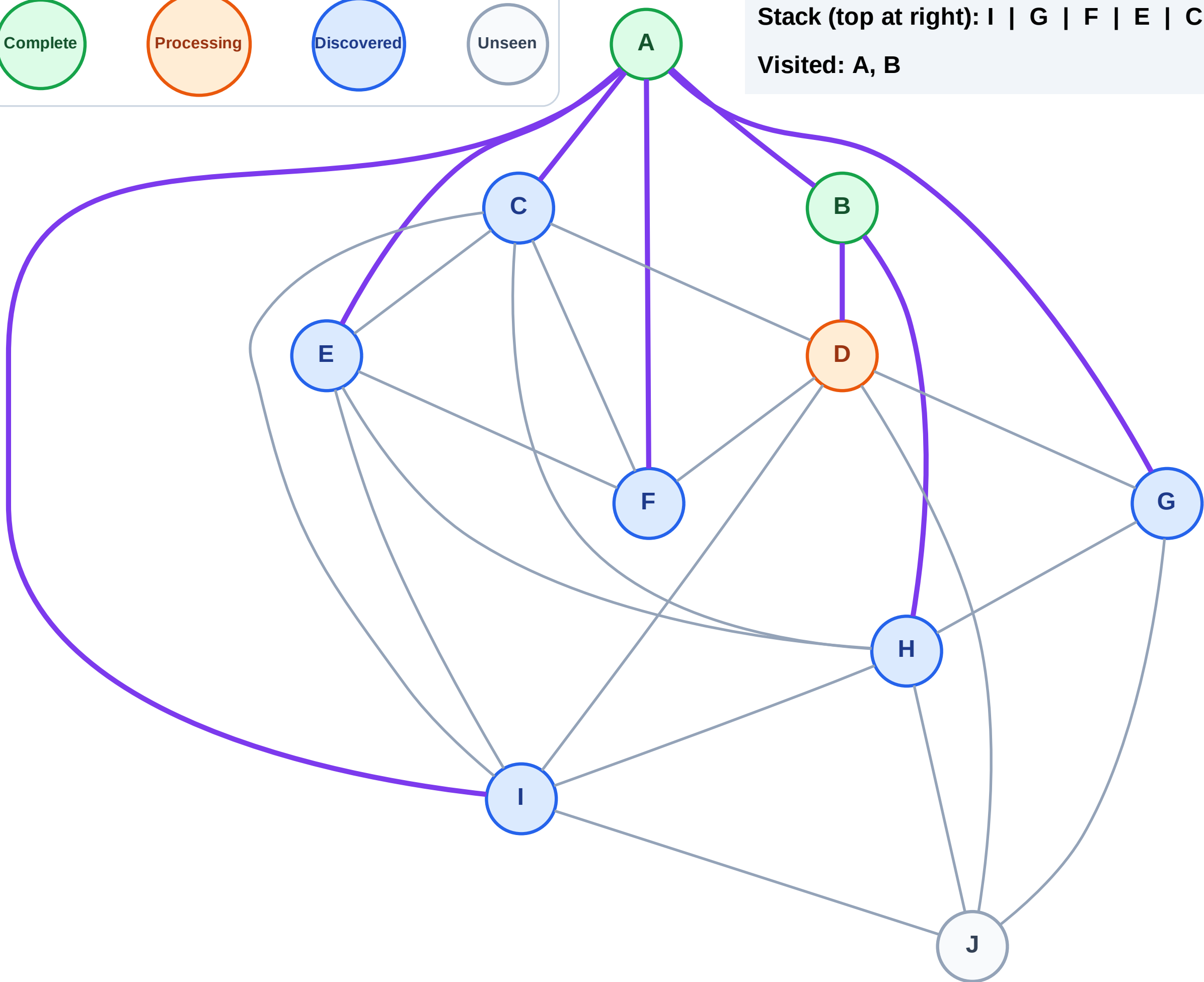
Step 6: Process node D

## Legend



Stack (top at right): I | G | F | E | C | H

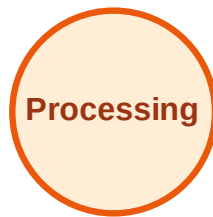
Visited: A, B



# DFS Traversal

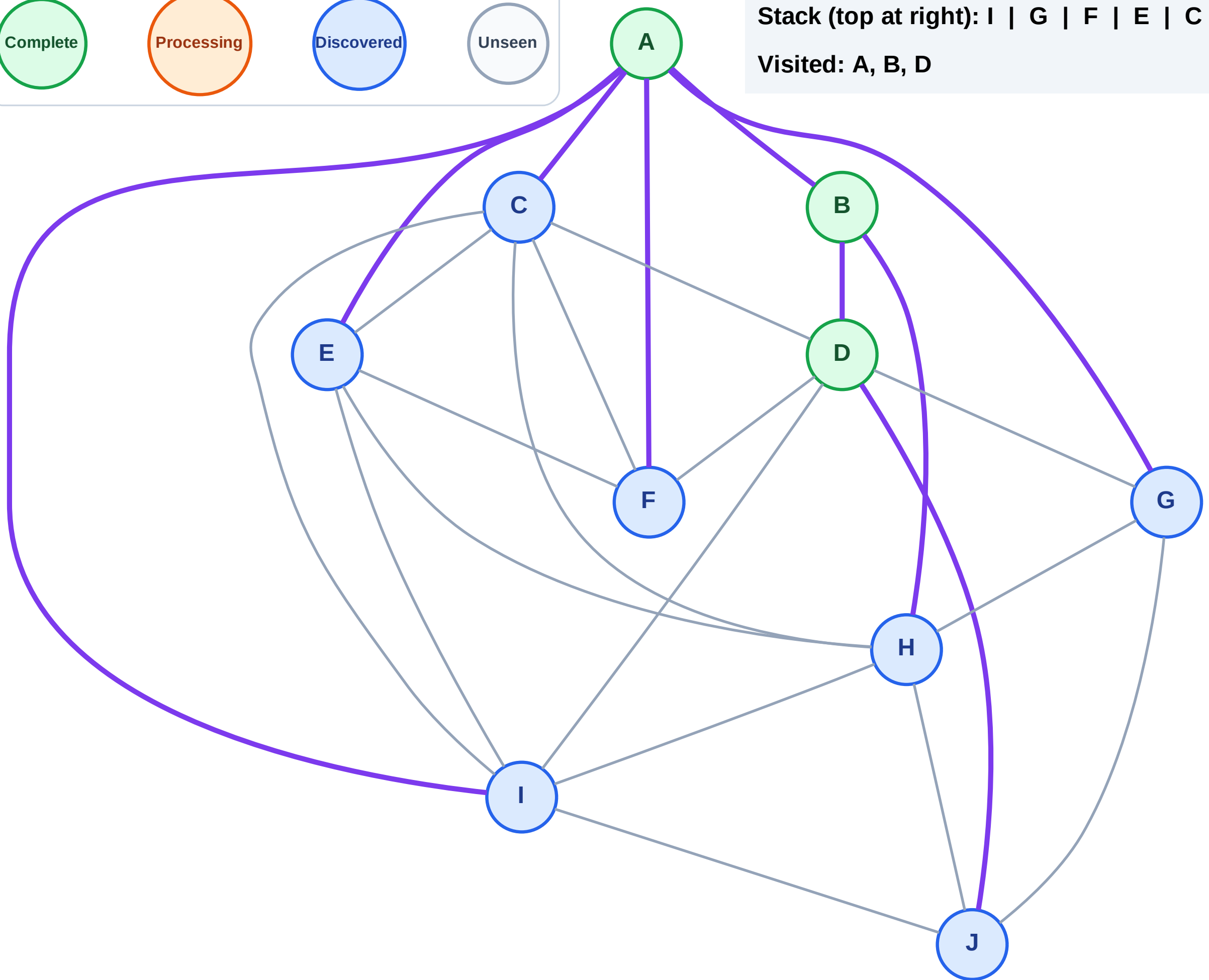
Step 7: Discover J from D

## Legend



Stack (top at right): I | G | F | E | C | H | J

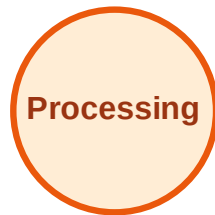
Visited: A, B, D



# DFS Traversal

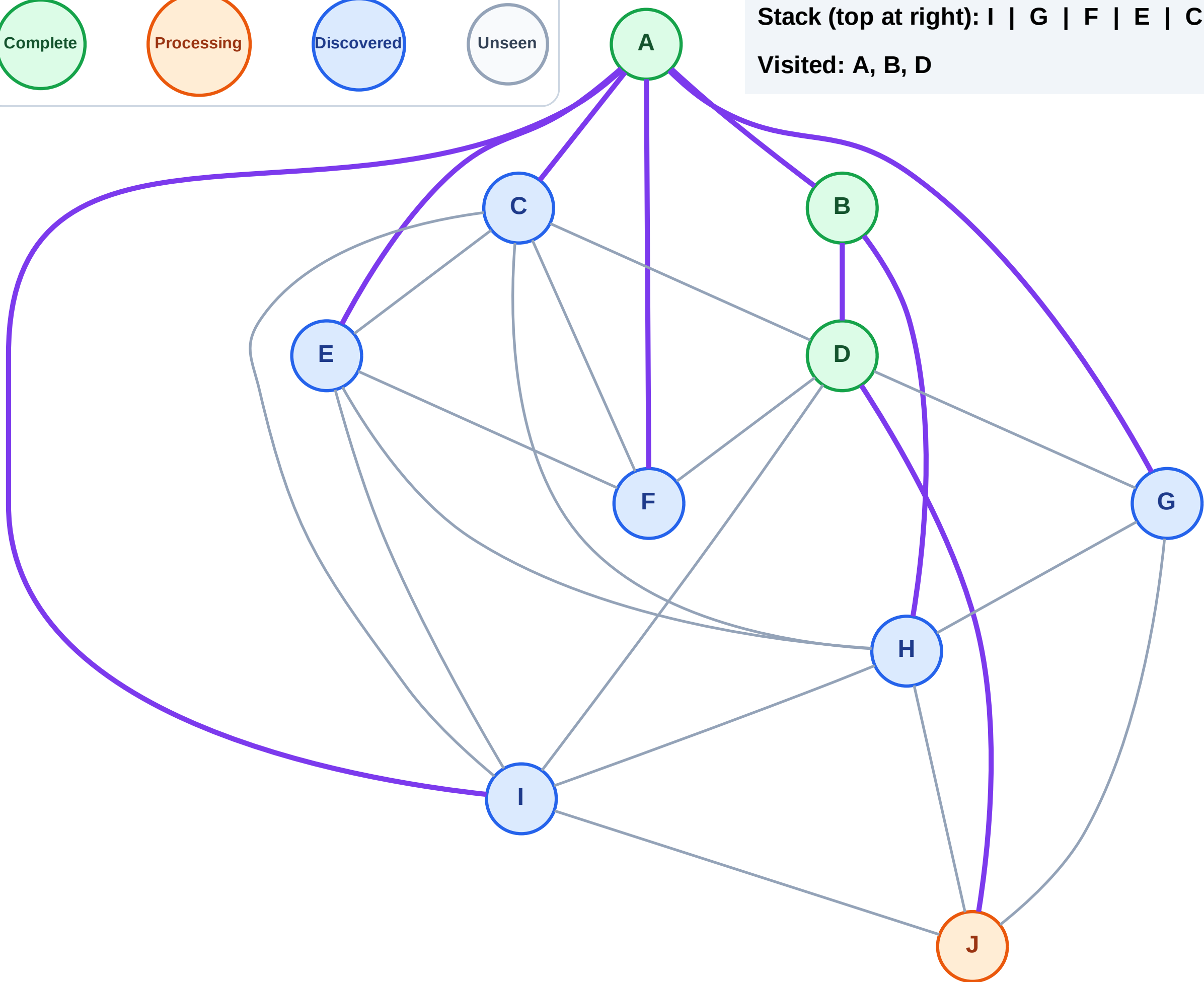
Step 8: Process node J

## Legend



Stack (top at right): I | G | F | E | C | H

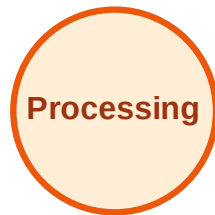
Visited: A, B, D



# DFS Traversal

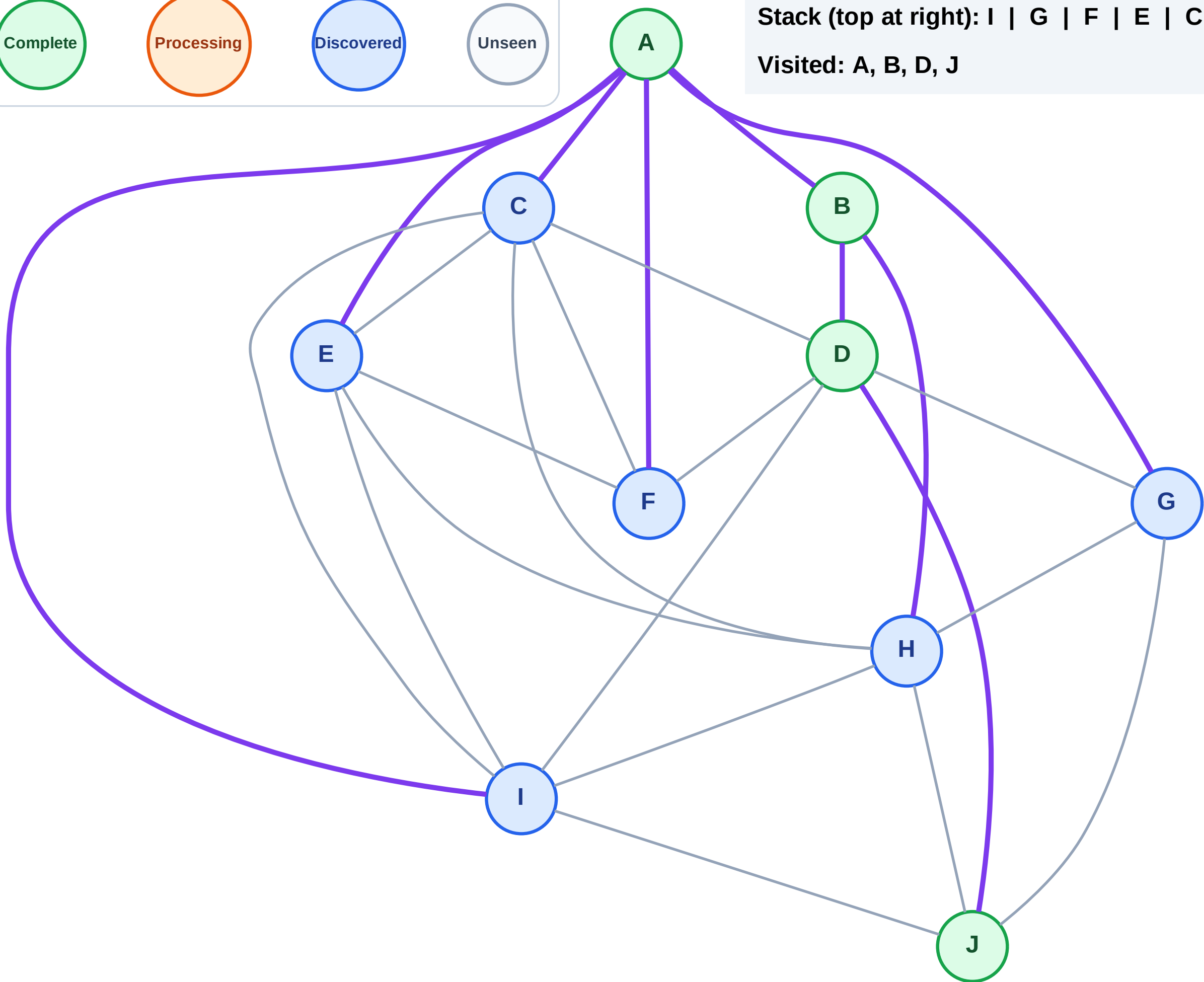
Step 9: No new neighbors from J

## Legend



Stack (top at right): I | G | F | E | C | H

Visited: A, B, D, J

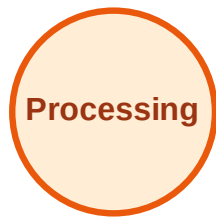




# DFS Traversal

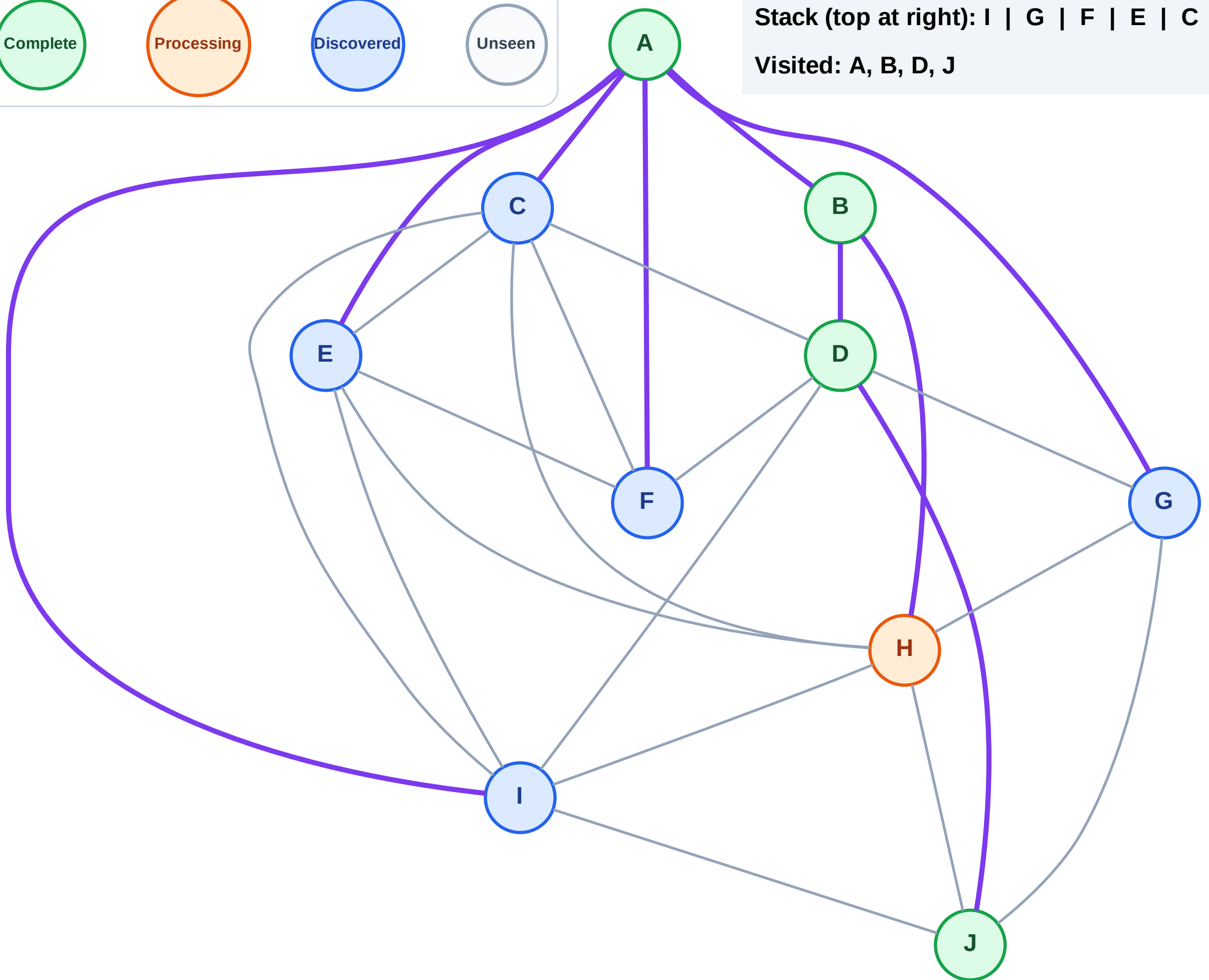
Step 10: Process node H

## Legend



Stack (top at right): I | G | F | E | C

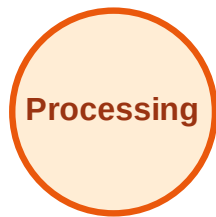
Visited: A, B, D, J



# DFS Traversal

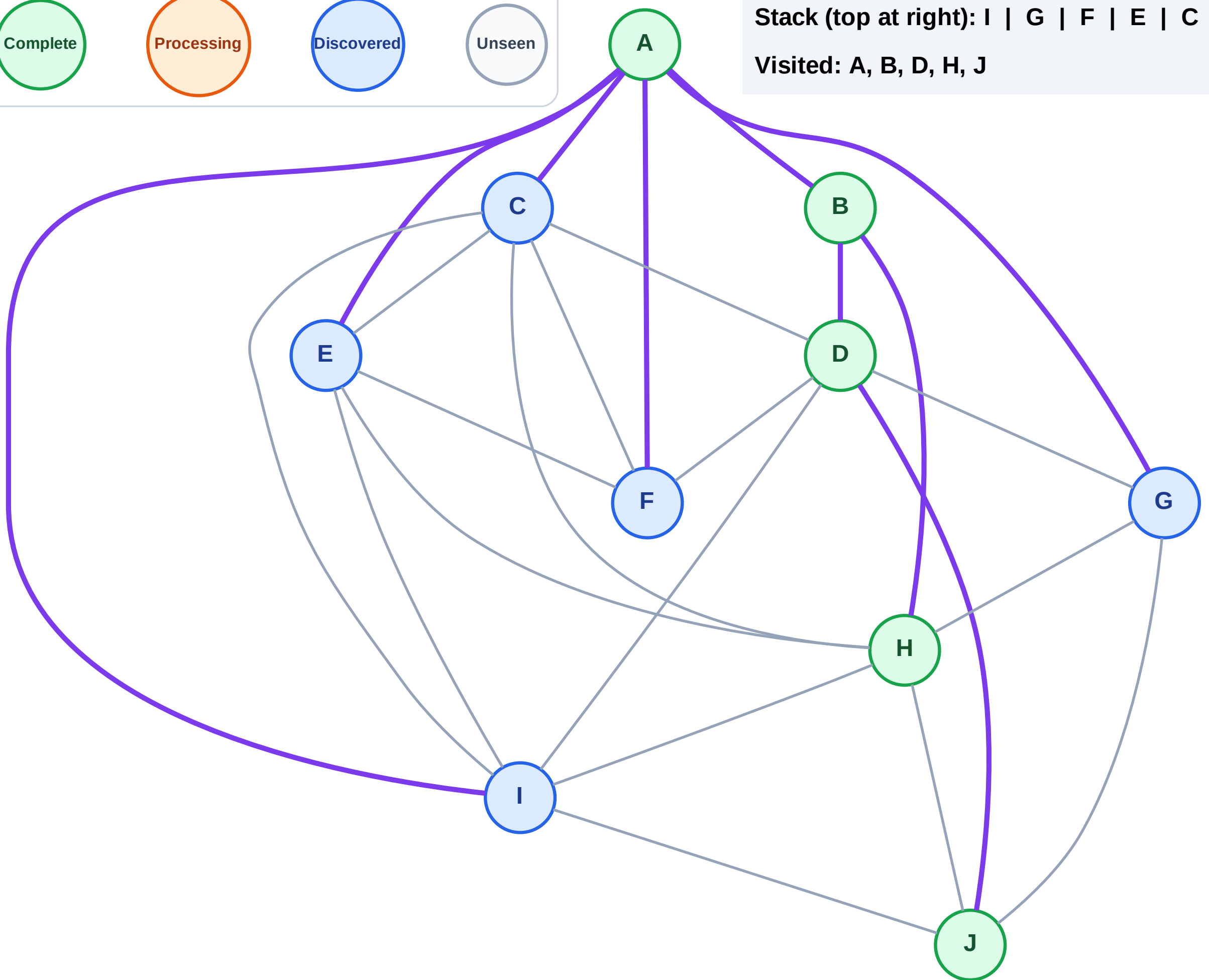
Step 11: No new neighbors from H

## Legend



Stack (top at right): I | G | F | E | C

Visited: A, B, D, H, J

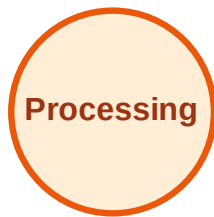




# DFS Traversal

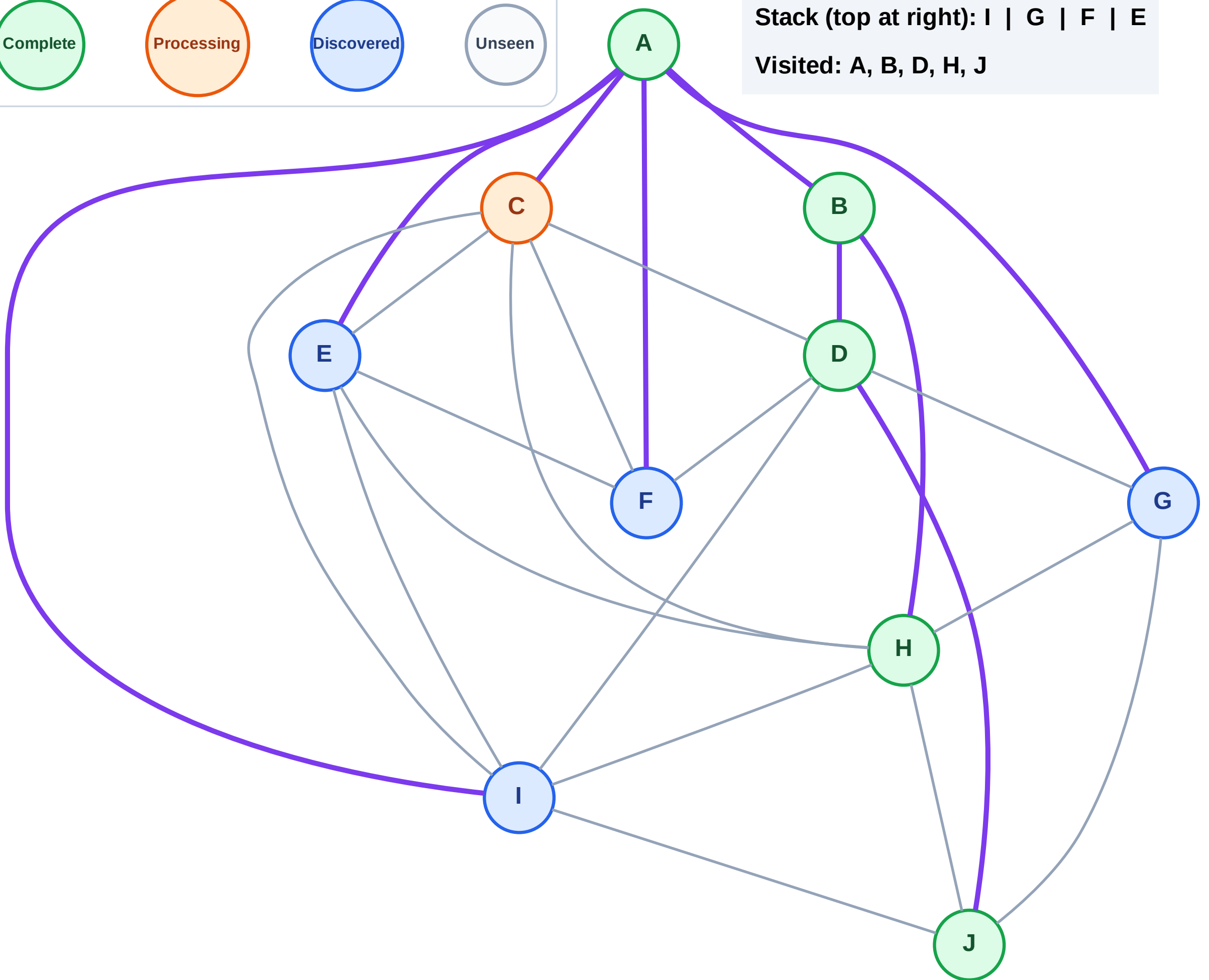
Step 12: Process node C

## Legend



Stack (top at right): I | G | F | E

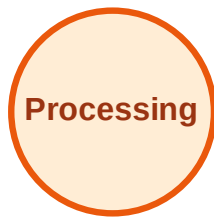
Visited: A, B, D, H, J



# DFS Traversal

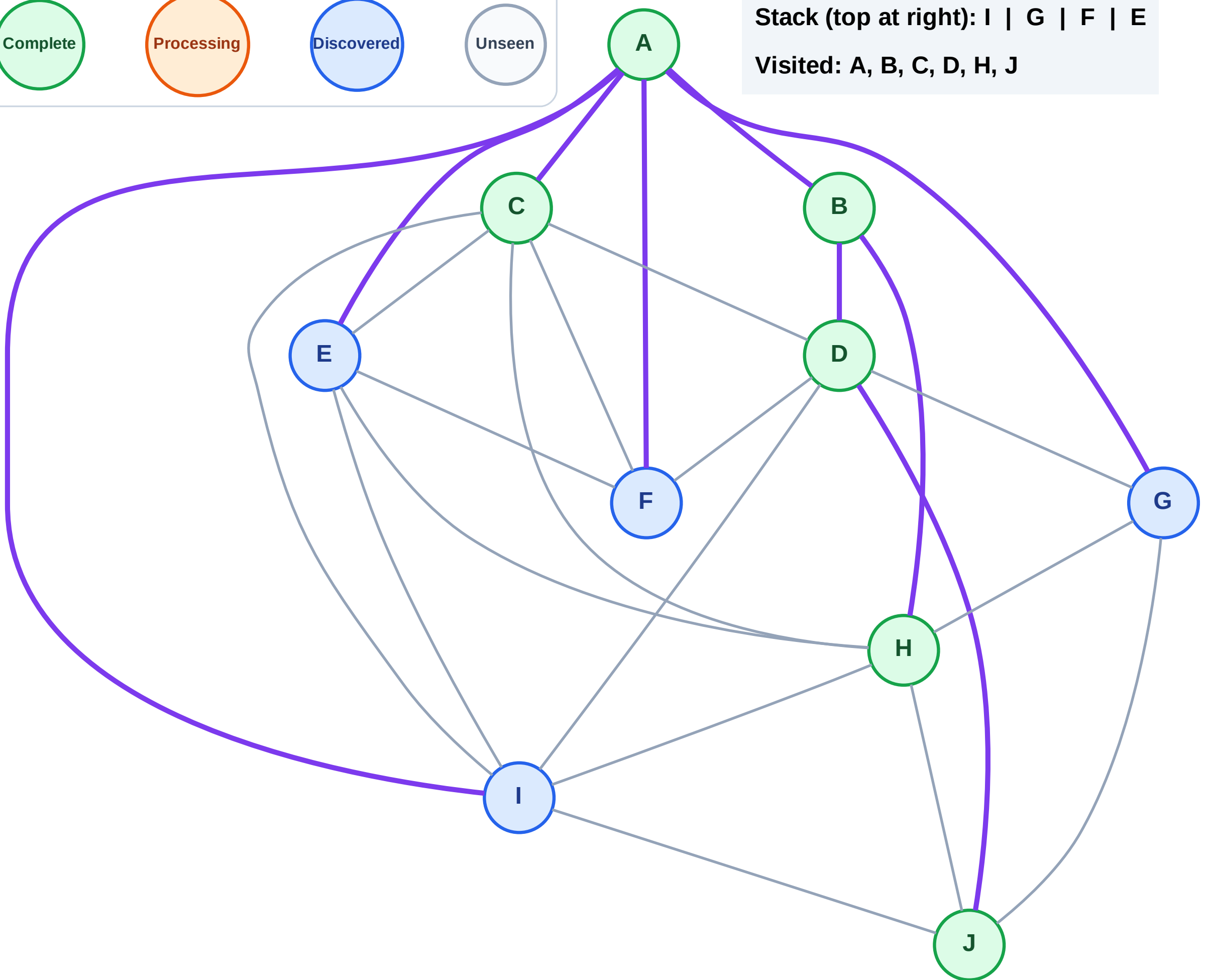
Step 13: No new neighbors from C

## Legend



Stack (top at right): I | G | F | E

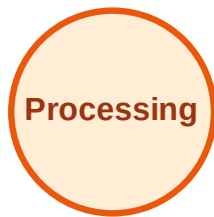
Visited: A, B, C, D, H, J



# DFS Traversal

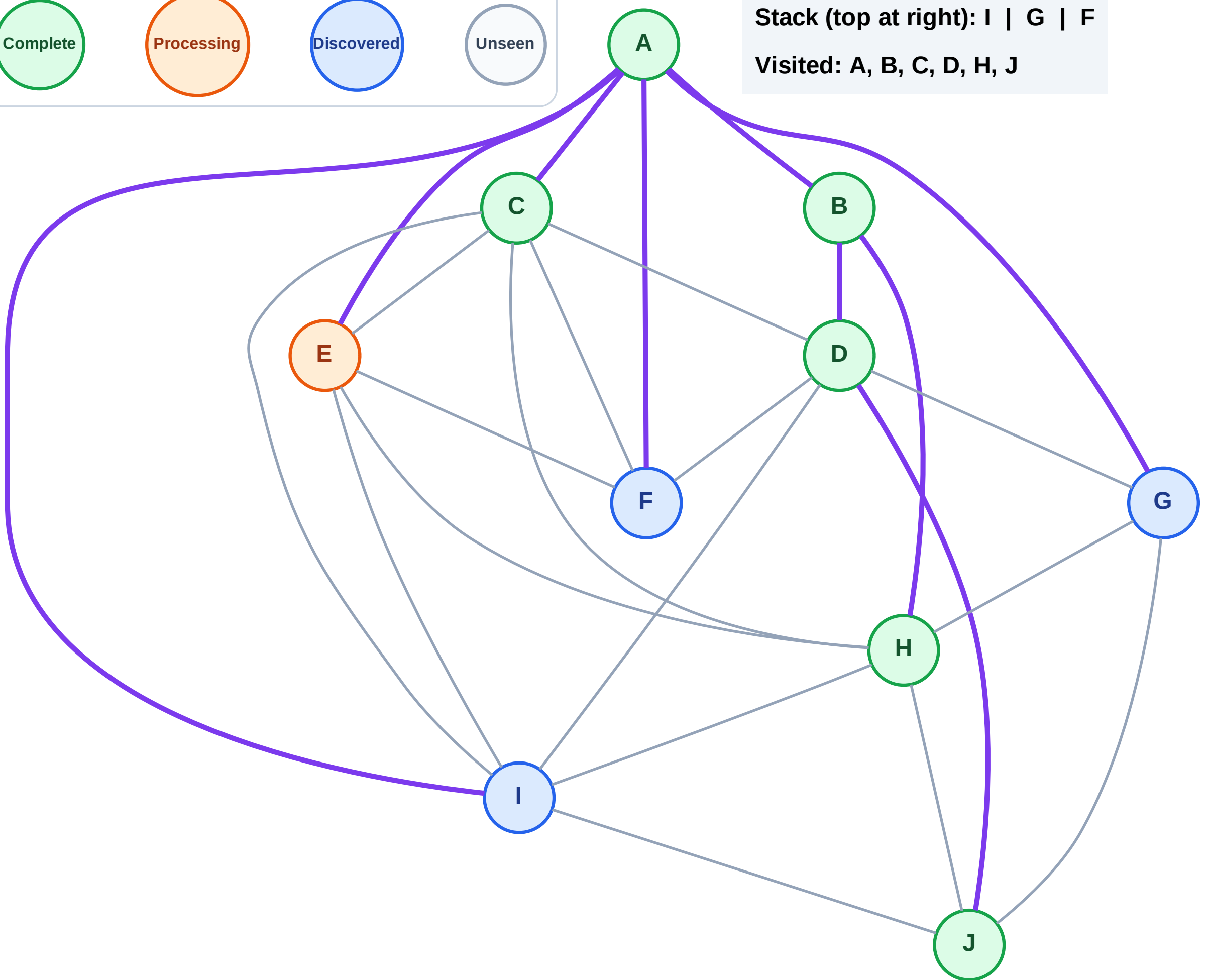
Step 14: Process node E

## Legend



Stack (top at right): I | G | F

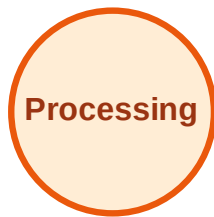
Visited: A, B, C, D, H, J



# DFS Traversal

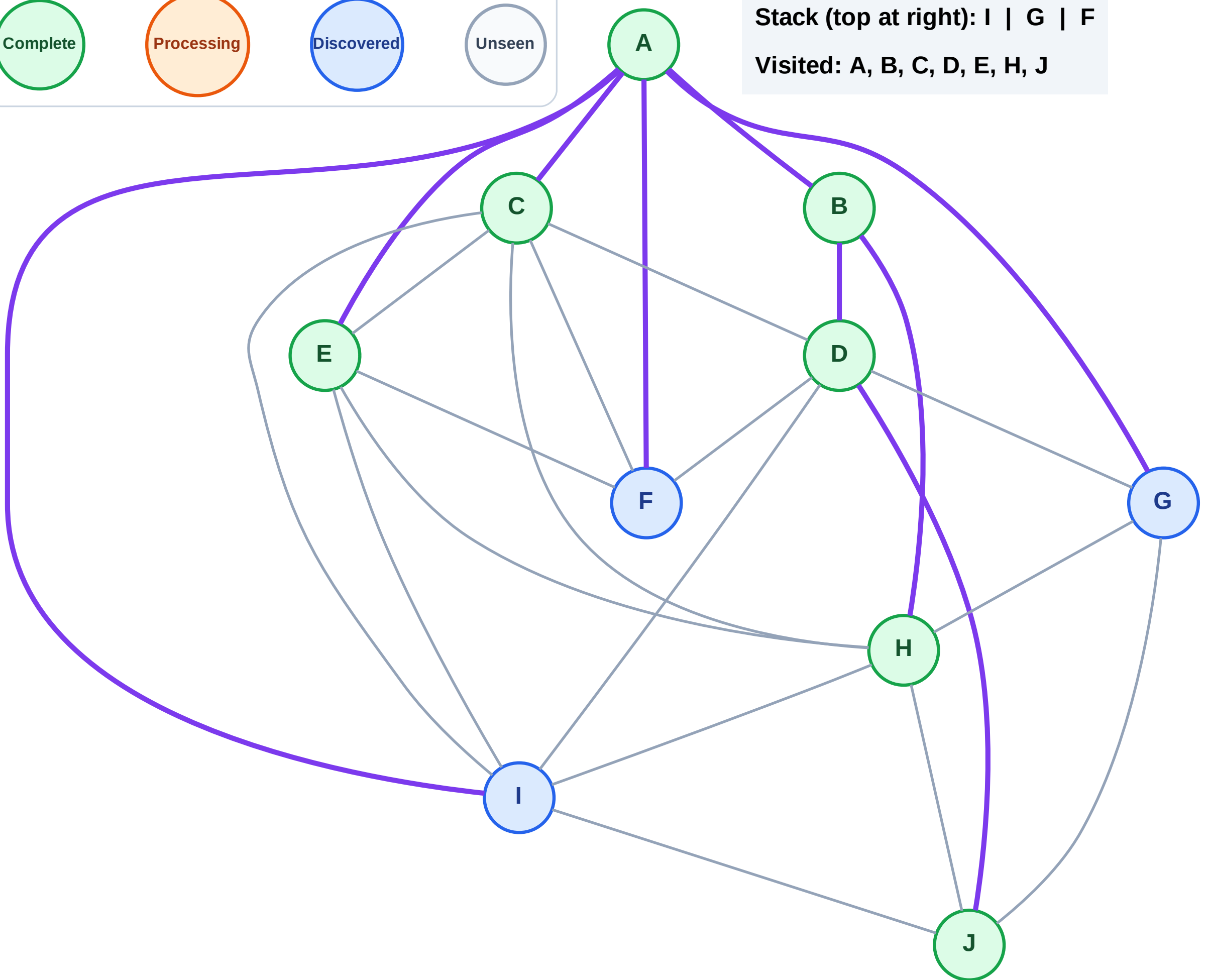
Step 15: No new neighbors from E

## Legend



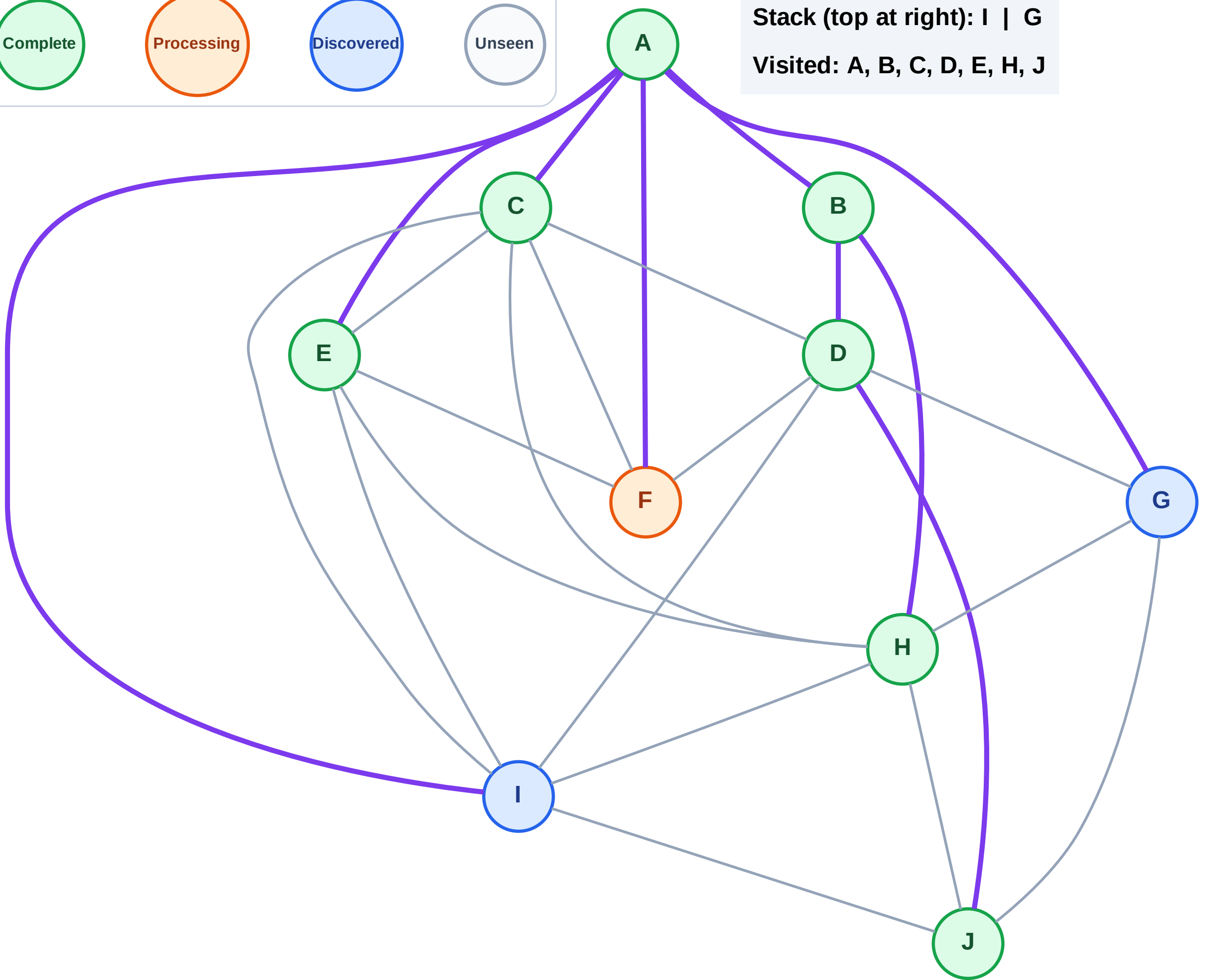
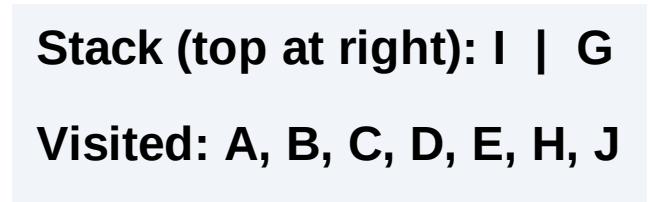
Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



### Step 16: Process node F

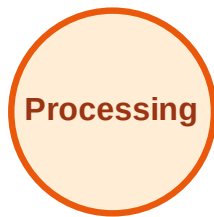
\_\_\_\_\_



# DFS Traversal

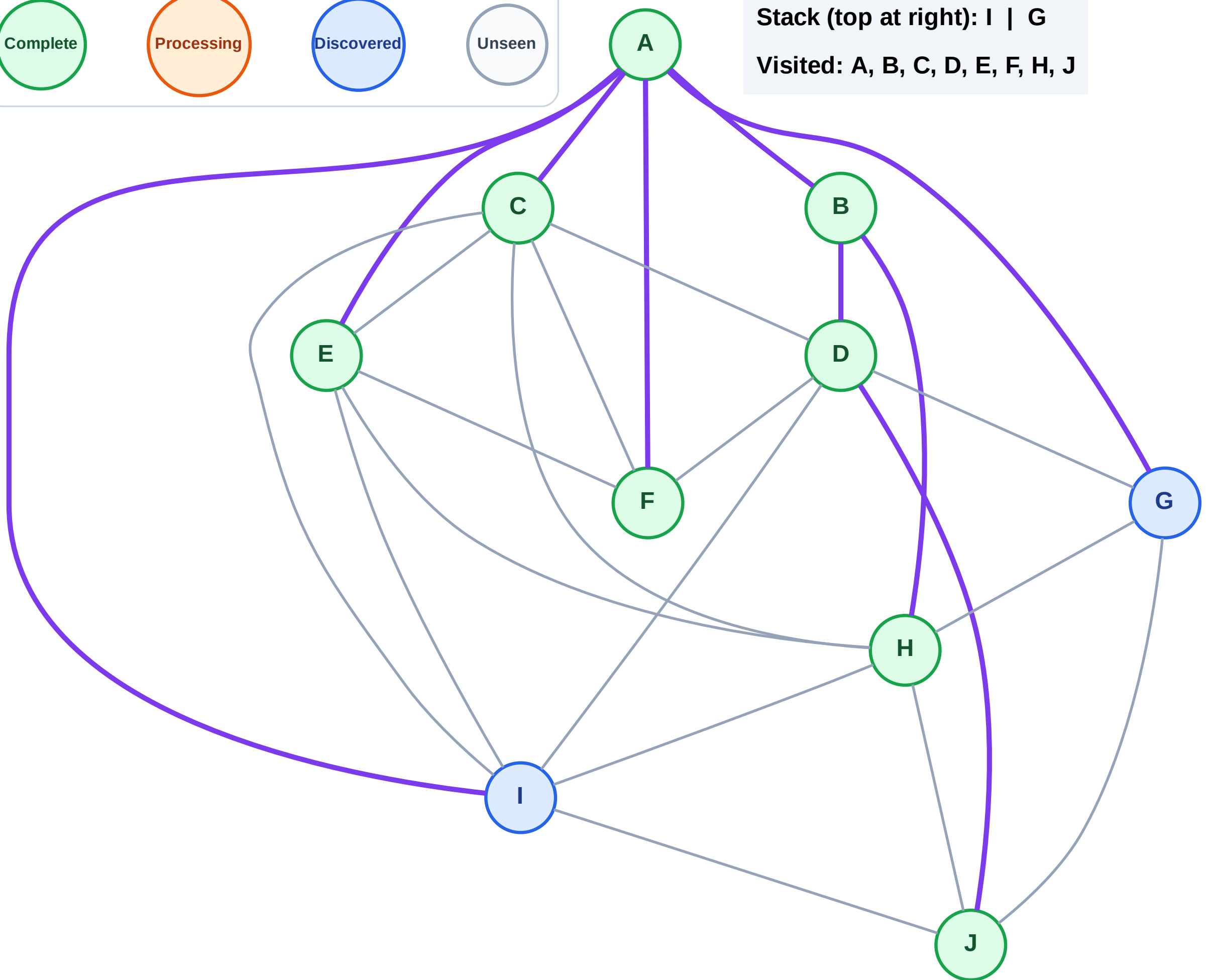
Step 17: No new neighbors from F

## Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J

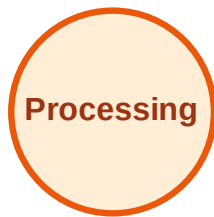




# DFS Traversal

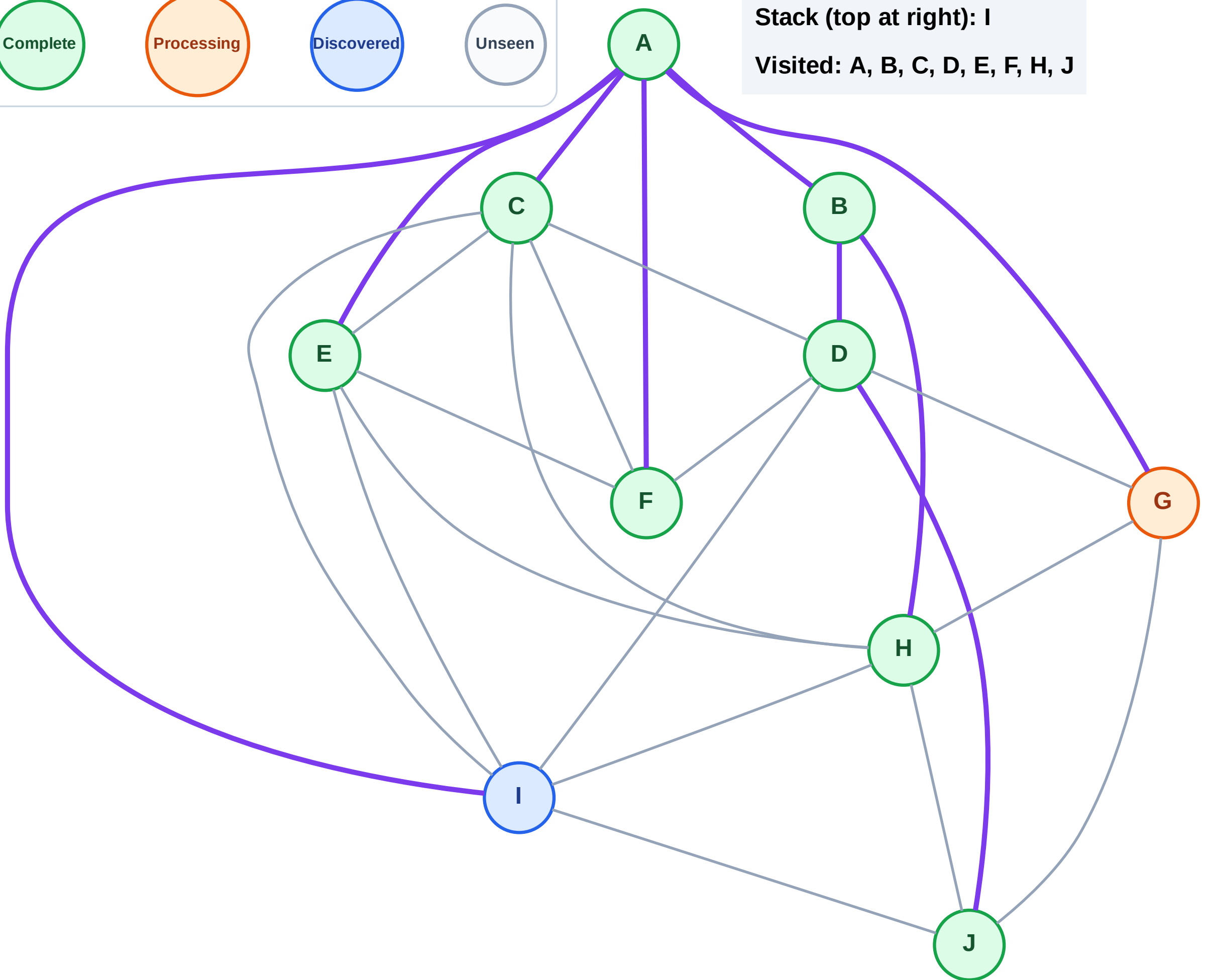
Step 18: Process node G

## Legend



Stack (top at right): I

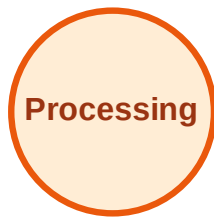
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

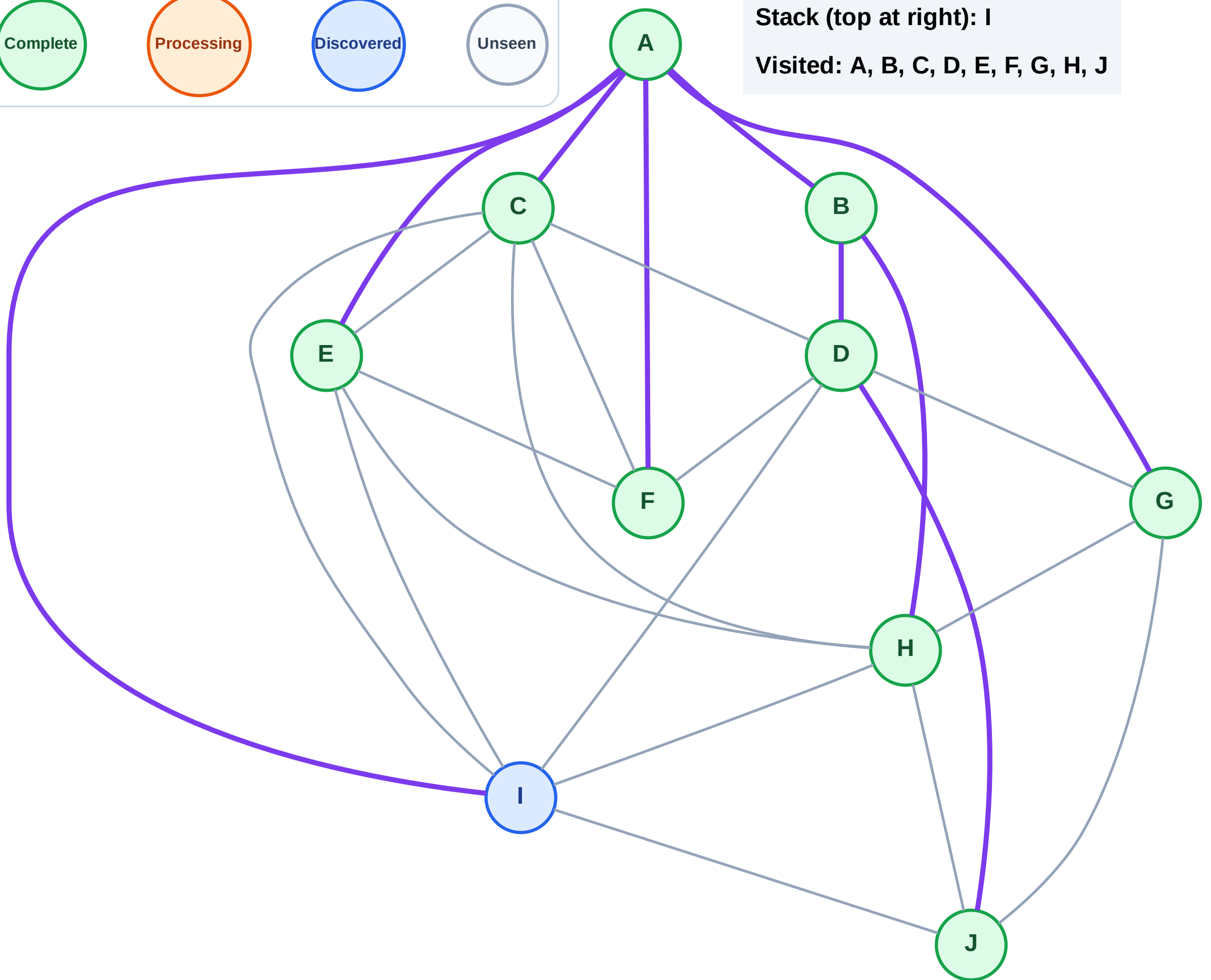
Step 19: No new neighbors from G

## Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

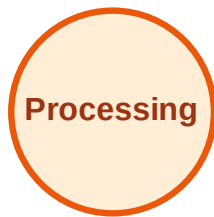




# DFS Traversal

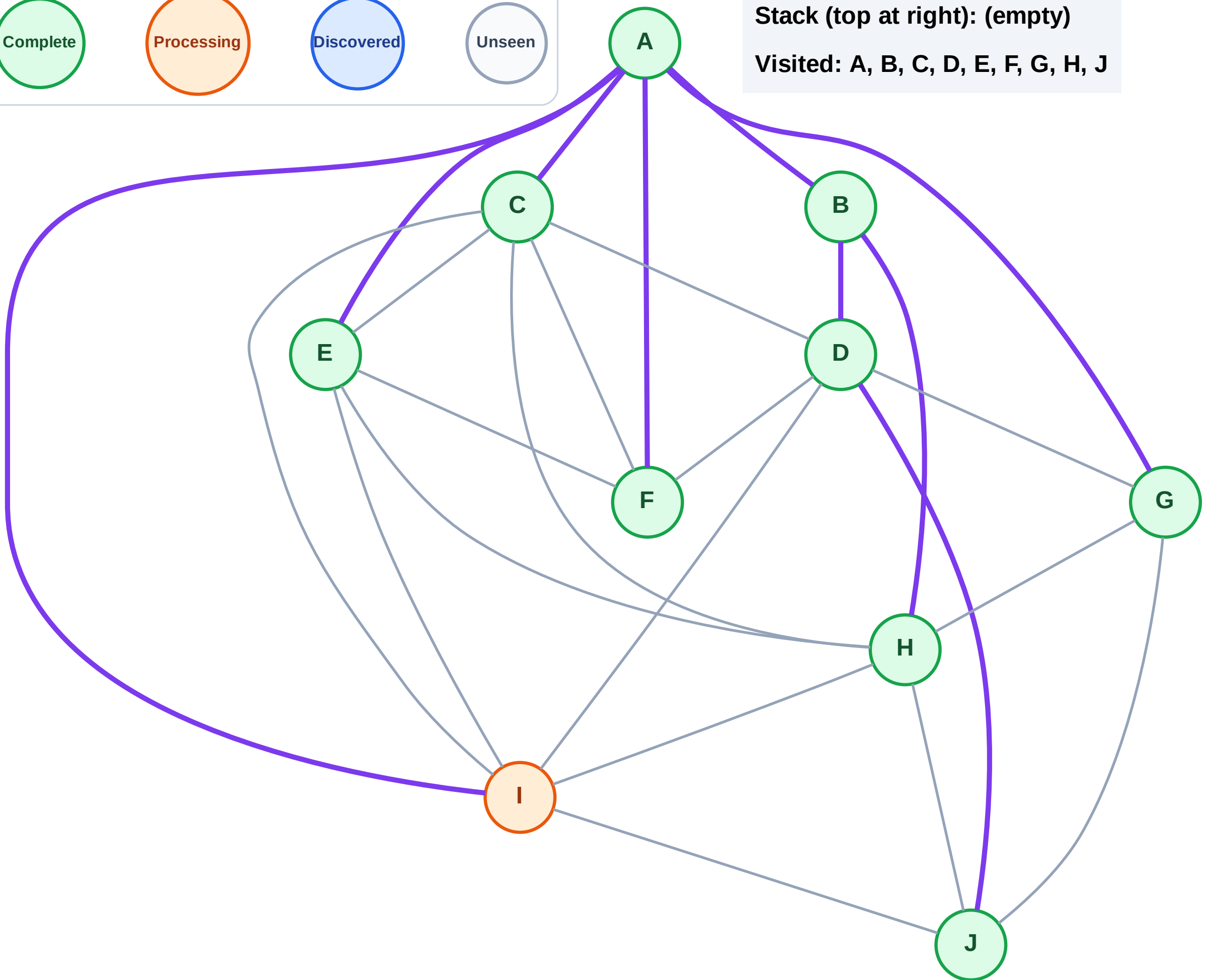
Step 20: Process node I

## Legend



Stack (top at right): (empty)

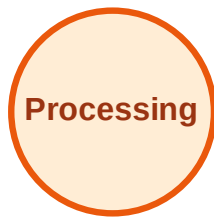
Visited: A, B, C, D, E, F, G, H, J



# DFS Traversal

Step 21: No new neighbors from I

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

